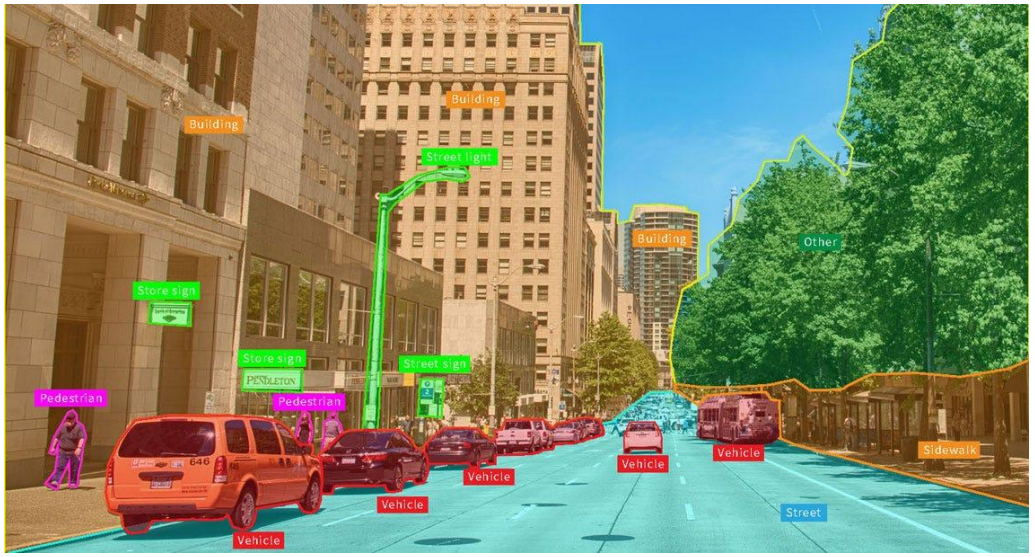
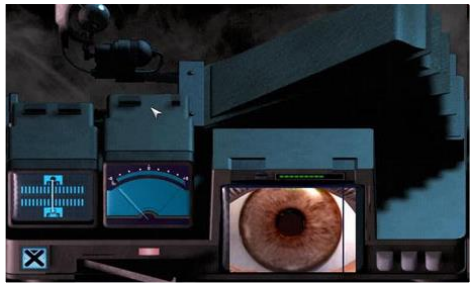
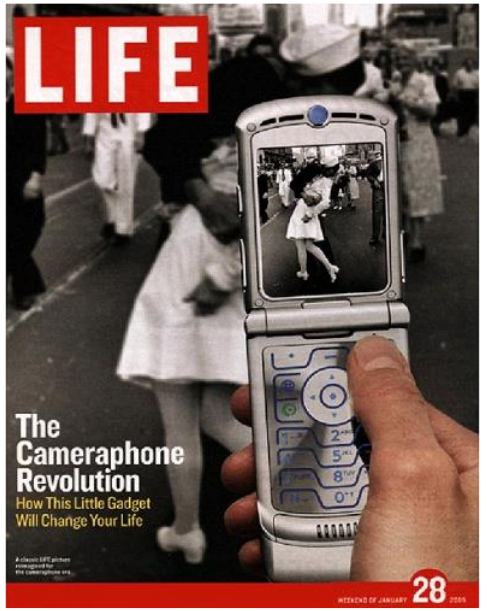
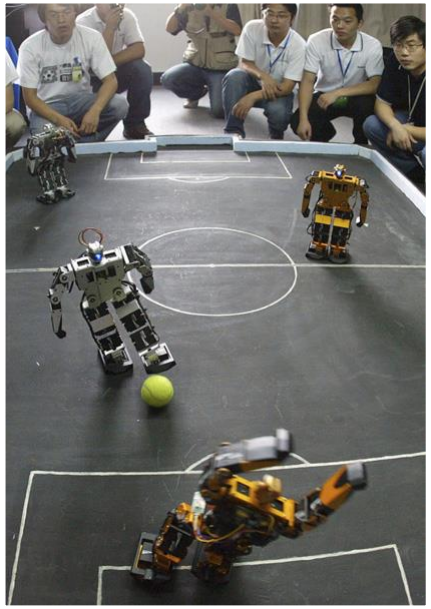
Welcome to computer vision !

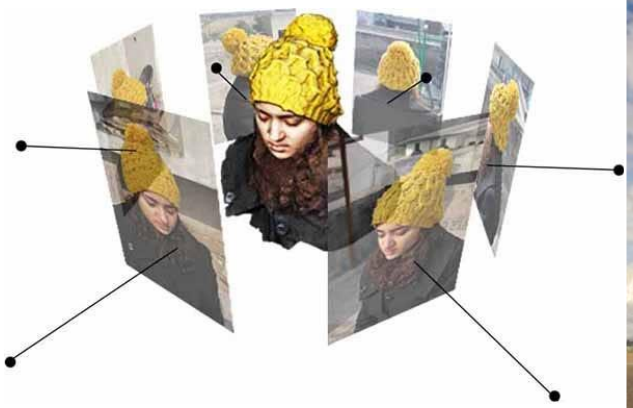


















What is vision ?

Bridging the gap between "Pixels" and "Meaning"



What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a camera sees

Vision is about understanding the world from visual input

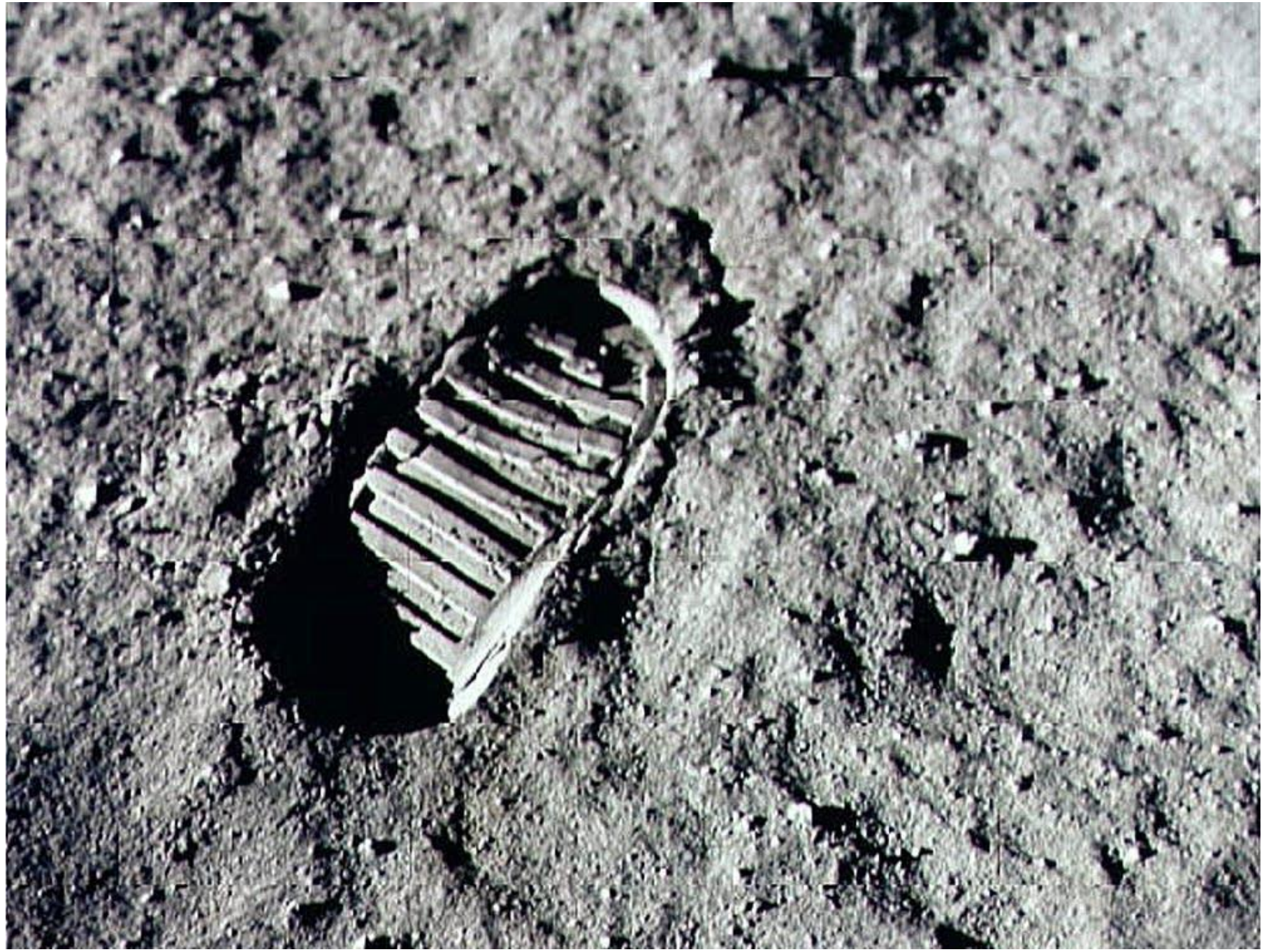
What is happening in the scene?





What objects are present in the scene?

White a short story about what happens in the scene?



Same pixels, different stories





What do we see in this image?



What is Vision?

Our Mind creates Vision

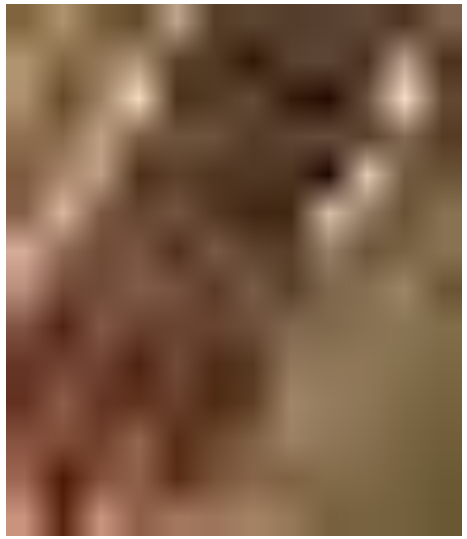
Vision puts "Things" into larger contextual "Stories" to give them "Meaning"

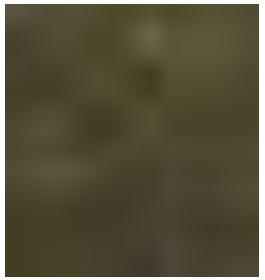


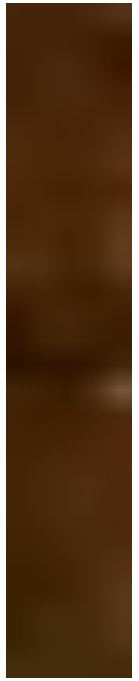
Pixels



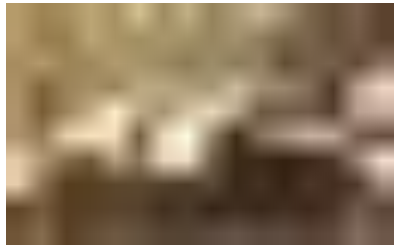
What is shown in these images?





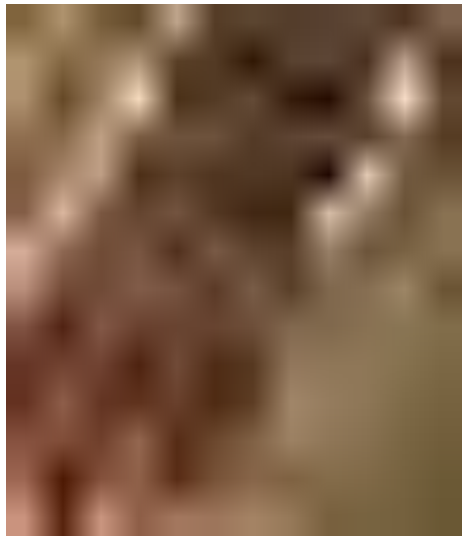




What is shown in these images?

Hand



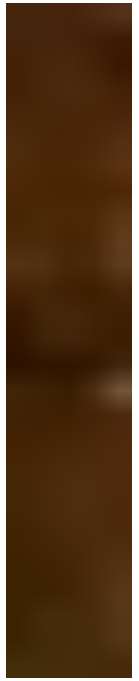
Shoe



Leg



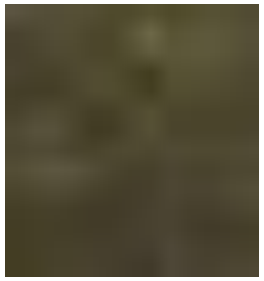
Tree



Leg



Leaves



Human face



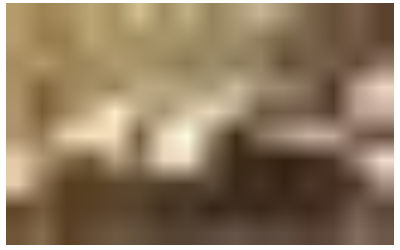
Branches with leaves



Tent



Food



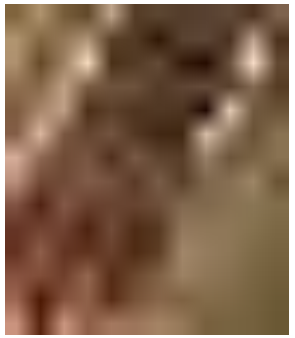
What does it mean to "see"?

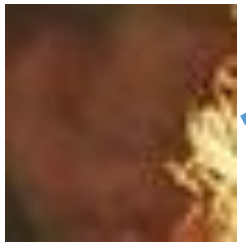


Leg

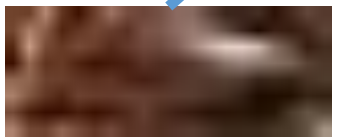


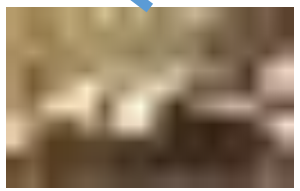
Hand





Shoe





Food

Who is cooking?

„**A lady** teaches some children how to prepare food for the next day. They are all getting ready to spend the night in the tent. It seems that the kids are in a camp, where they learn how to survive in nature.”

„A group of 4 people are in the woods, on the shore of a lake. I believe it is a family on vacation or on a trip. **The father** shows the others how to cook on the grill and they look and give their opinion on what he is doing there.”

„A family is camping. The family consists of 3 children and **a mother**. They are in a forest near a lake. The forest looks reddish because of the sun at sunset. The mother prepares something to eat while the children look at her. Next to them is a tent. "

„The image represents a natural scenery of a forest. Where **an adult blonde caucasian female** prepares a grill for 3 children of different ages and race. They are camping near a lake in the afternoon sun. Weather appears to be warm, probably during summer.”

Different descriptions given independently by my AI Master students, POLITEHNICA Bucharest

What are the children doing?

„A lady teaches some children how to prepare food for the next day. They are all getting ready to spend the night in the tent. It seems that the kids are in a camp, where **they learn how to survive in nature**."

„A group of 4 people are in the woods, on the shore of a lake. I believe it is a family on vacation or on a trip. The father shows the others how to cook on the grill and they look and **give their opinion on what he is doing there**."

„A family is camping. The family consists of 3 children and a mother. They are in a forest near a lake. The forest looks reddish because of the sun at sunset. The mother prepares something to eat **while the children look at her**. Next to them is a tent. "

„The image represents a natural scenery of a forest. Where an adult blonde caucasian female prepares a grill for **3 children of different ages and race**. They are camping near a lake in the afternoon sun. Weather appears to be warm, probably during summer."

Different descriptions given independently by my AI Master students, POLITEHNICA Bucharest

When is it happening?

„A lady teaches some children how to prepare food for the next day. They are getting ready to **spend the night in the tent**. It seems that the kids are in a camp, where they learn how to survive in nature."

The image denotes a quiet atmosphere **at the day's dawn** on a sky without clouds.

A family with 3 children, or just a mother with the three children, spent a night in the tent in nature near a lake / river. **Now they've been awake for a while** and are cooking something on the grill.

The photo is the scene of a lake campsite in the middle of summer. **Judging by the position of the sun, it is afternoon**, and the characters are most likely involved in the preparation of the meal.

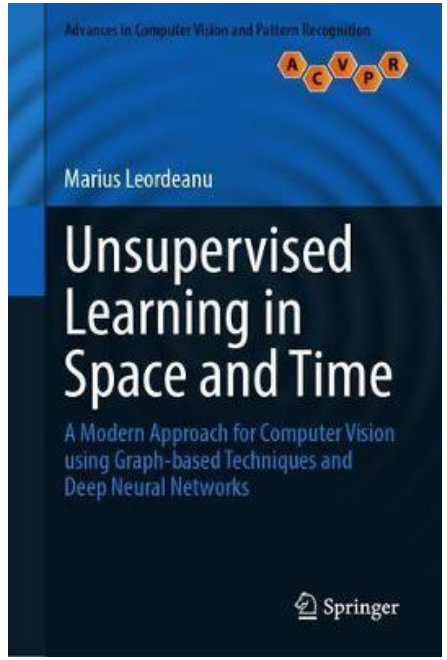
A family is camping. The family consists of 3 children and a mother. They are in a forest near a lake. The forest looks reddish **because of the sun at sunset.**

Different descriptions given independently by my AI Master students, POLITEHNICA Bucharest

What else?

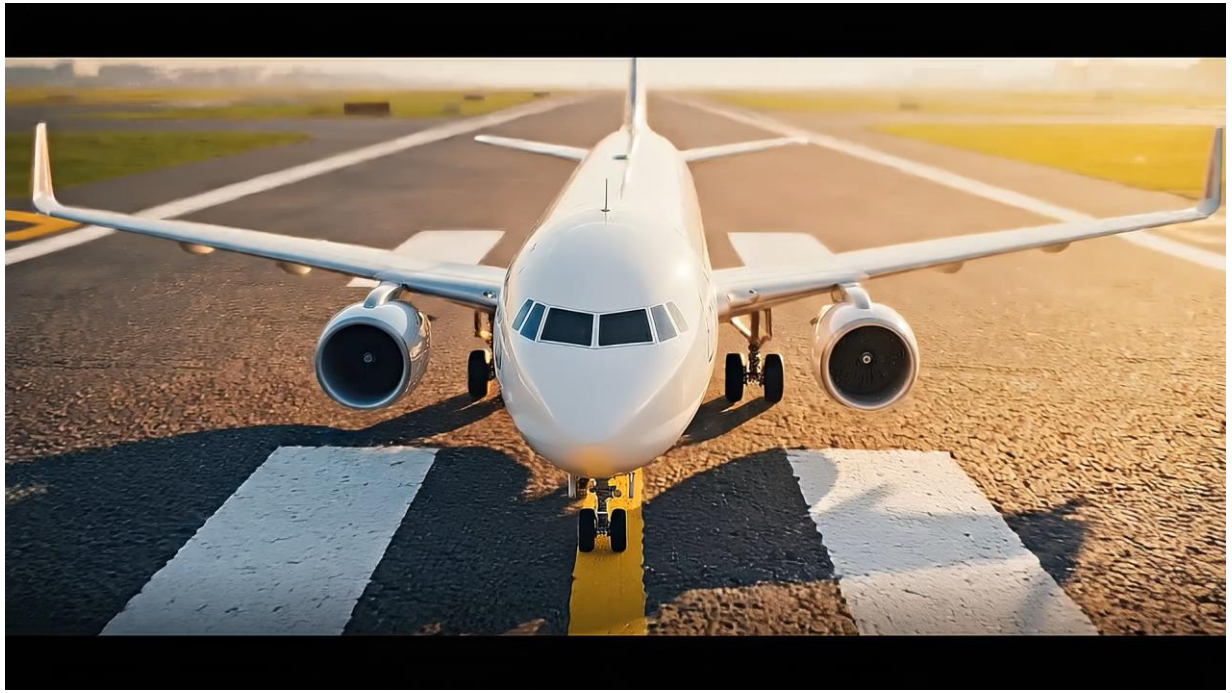
A family with 3 children, or just a mother with the three children, spent a night with the tent in the nature near a lake / river. Now, they've been awake for a while and are cooking something on the grill. Mother teaches kids how to take care of the barbecue. At the same time, she teaches them **to appreciate everything that happens ... the scent of burned wood, the beautiful light that surrounds them**. We have 4 people, a tent, a barbecue, fir trees, water, food, **grilling utensils**.

For more about the consensus between vision and language please check out my book.



Marius Leordeanu, Unsupervised Learning in Space and Time, Springer, 2020

Stories should respect Physics

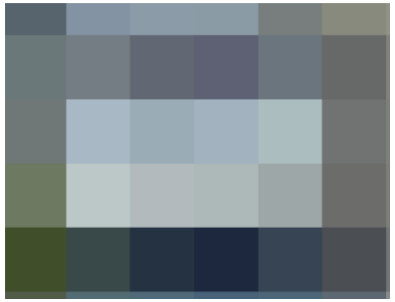




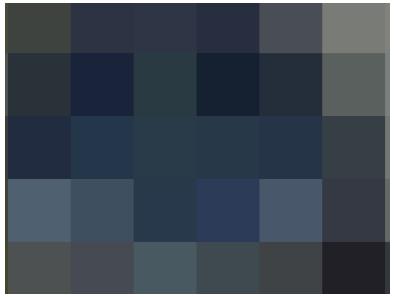
State-of-the Art Video Generation Models map language to vision without having an **understanding of a physical world model**

The role of visual context: tell me where you are and I tell you what you see









The role of visual context: tell me where you are and I tell you what you see



Optical Illusions

Show us that Vision is an Illusion
A creation of the mind based on guessing from visual input (pixels)

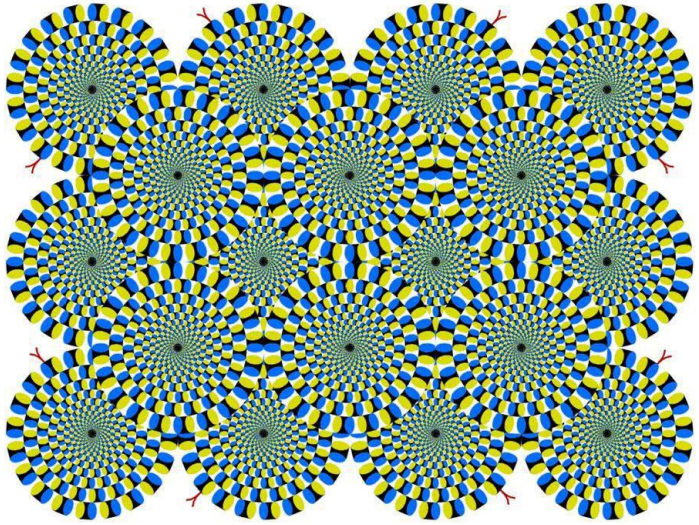
Perception is an inherently ambiguous problem

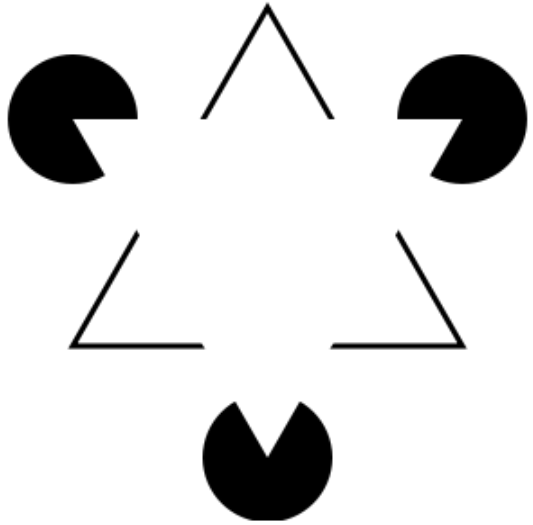
- Many different 3D scenes could have given rise to a particular 2D picture

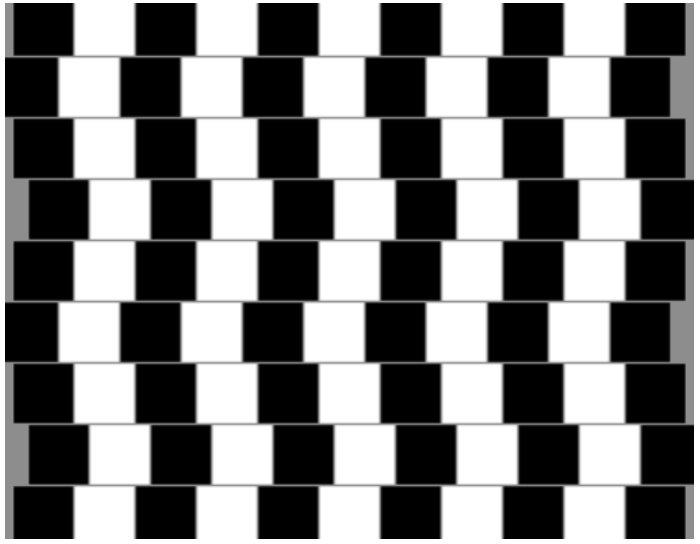


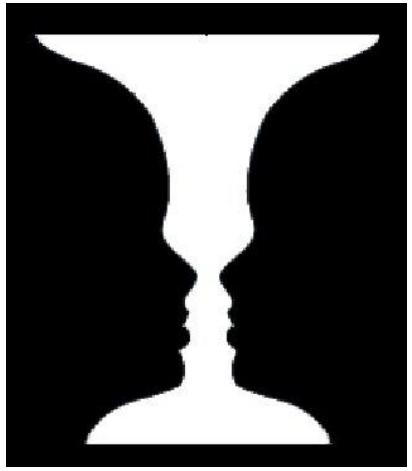


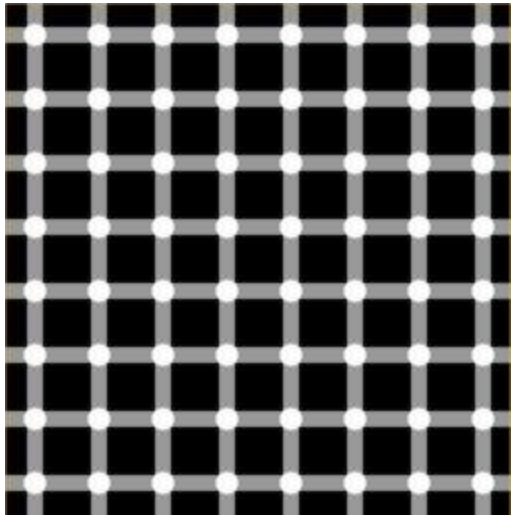
Is vision and illusion ?











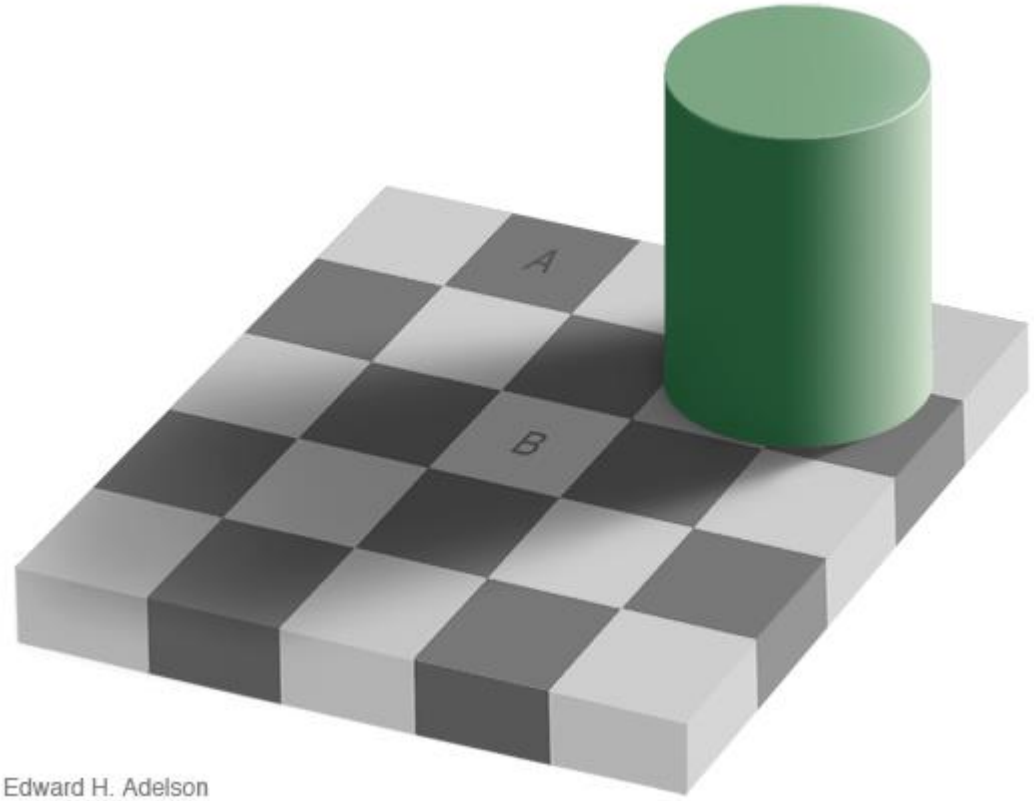
Chromatic adaptation



http://www.planetperplex.com/en/color_illusions.html

Slide credit: K. Grauman

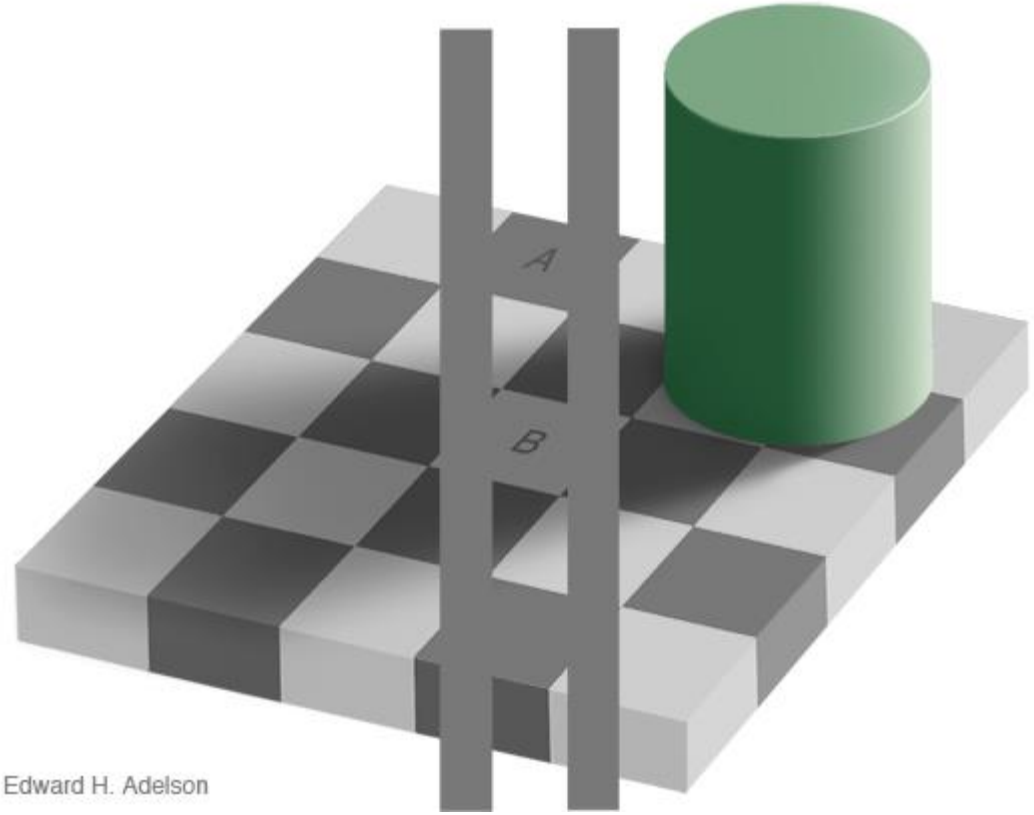
Checker shadow illusion



Edward H. Adelson

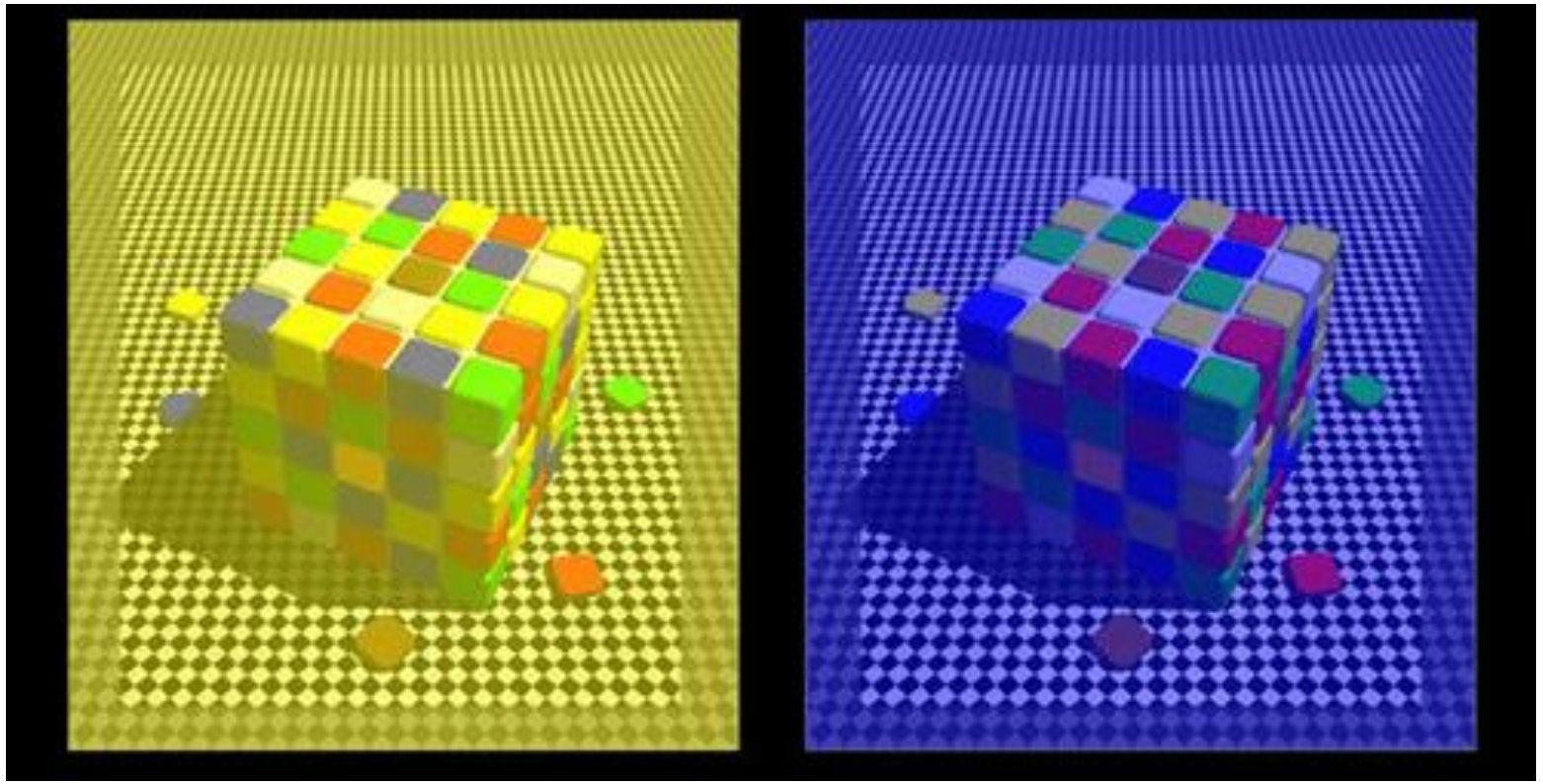
http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html

Checker shadow illusion



- Possible explanations
  - Simultaneous contrast
  - Reflectance edges vs. illumination edges

http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html

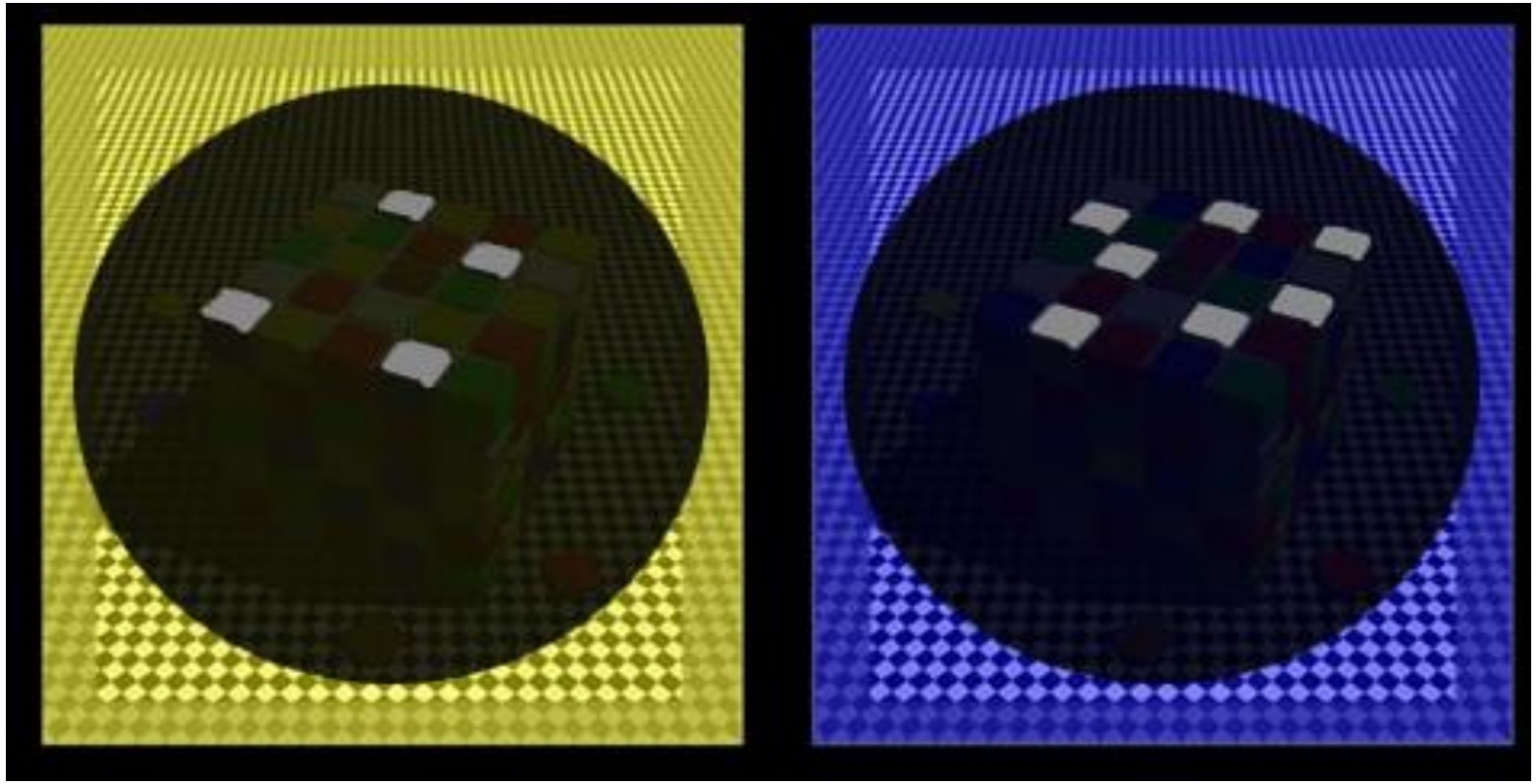


Look at blue squares

Look at yellow squares

- Content © 2008 R.Beau Lotto
- http://www.lottolab.org/articles/illusionsoflight.asp

Slide credit: K. Grauman

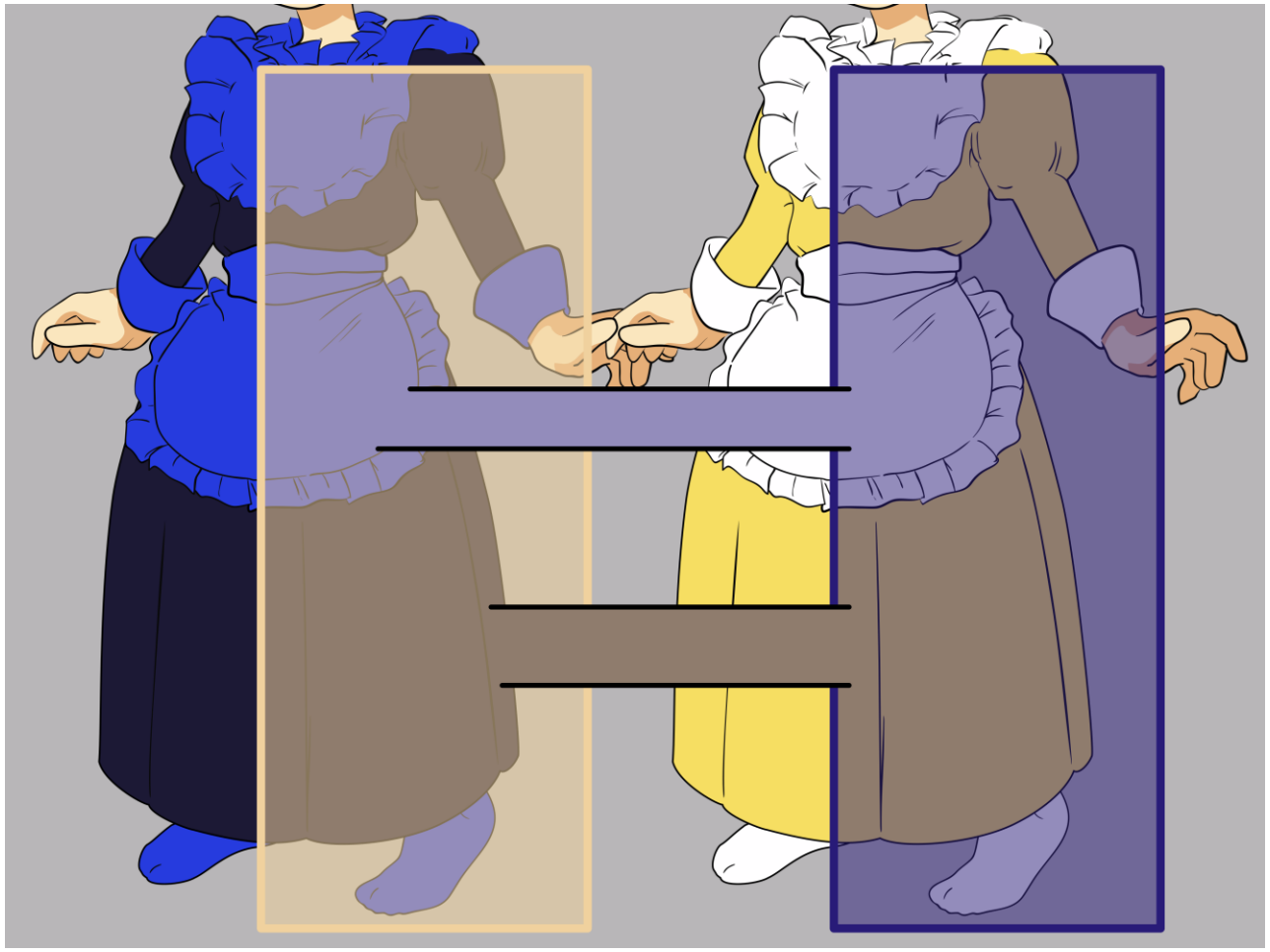


- Content © 2008 R.Beau Lotto
- http://www.lottolab.org/articles/illusionsoflight.asp

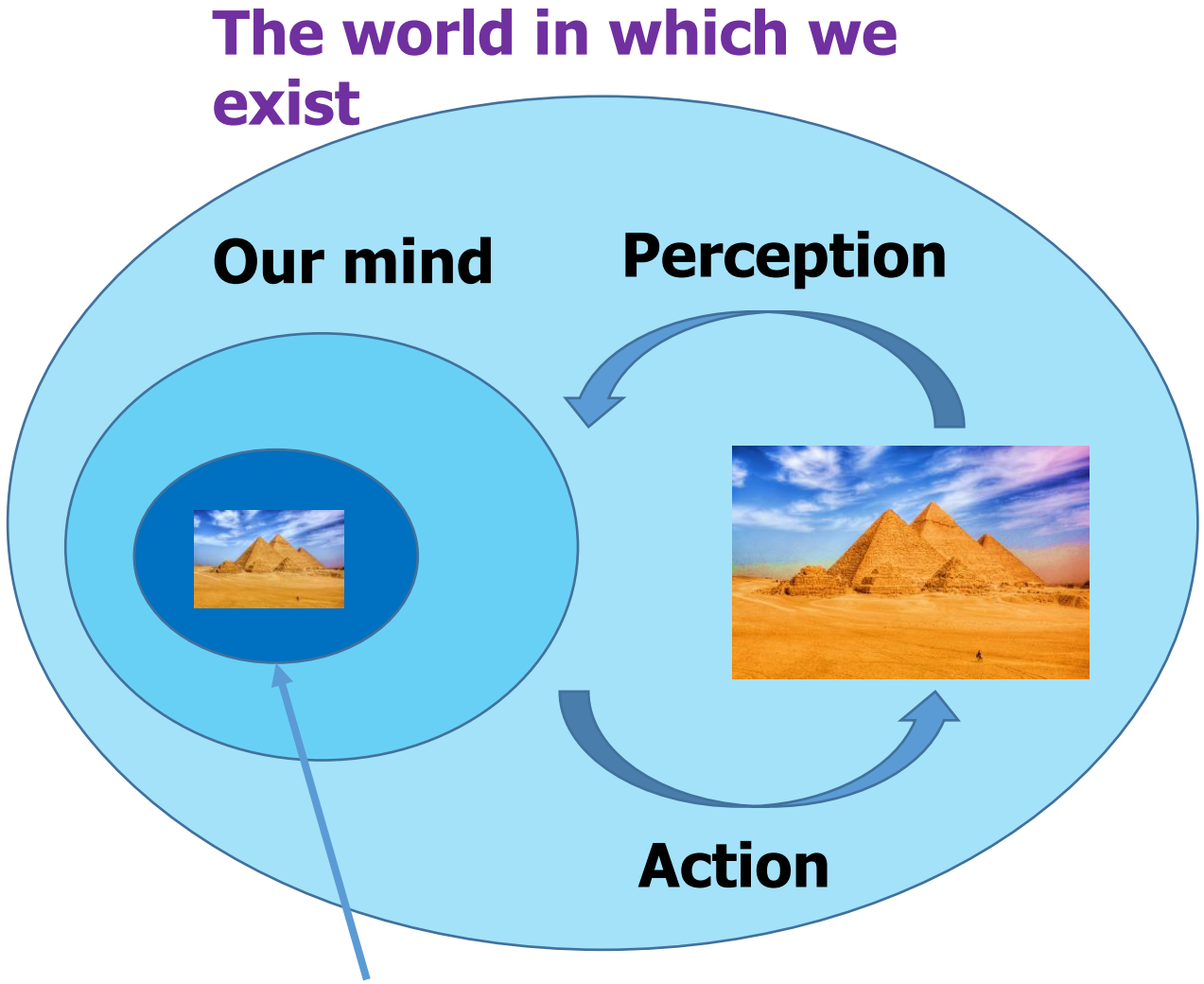
Slide credit: K. Grauman

The Dress Illusion





The Reality of Mind - The World and Us

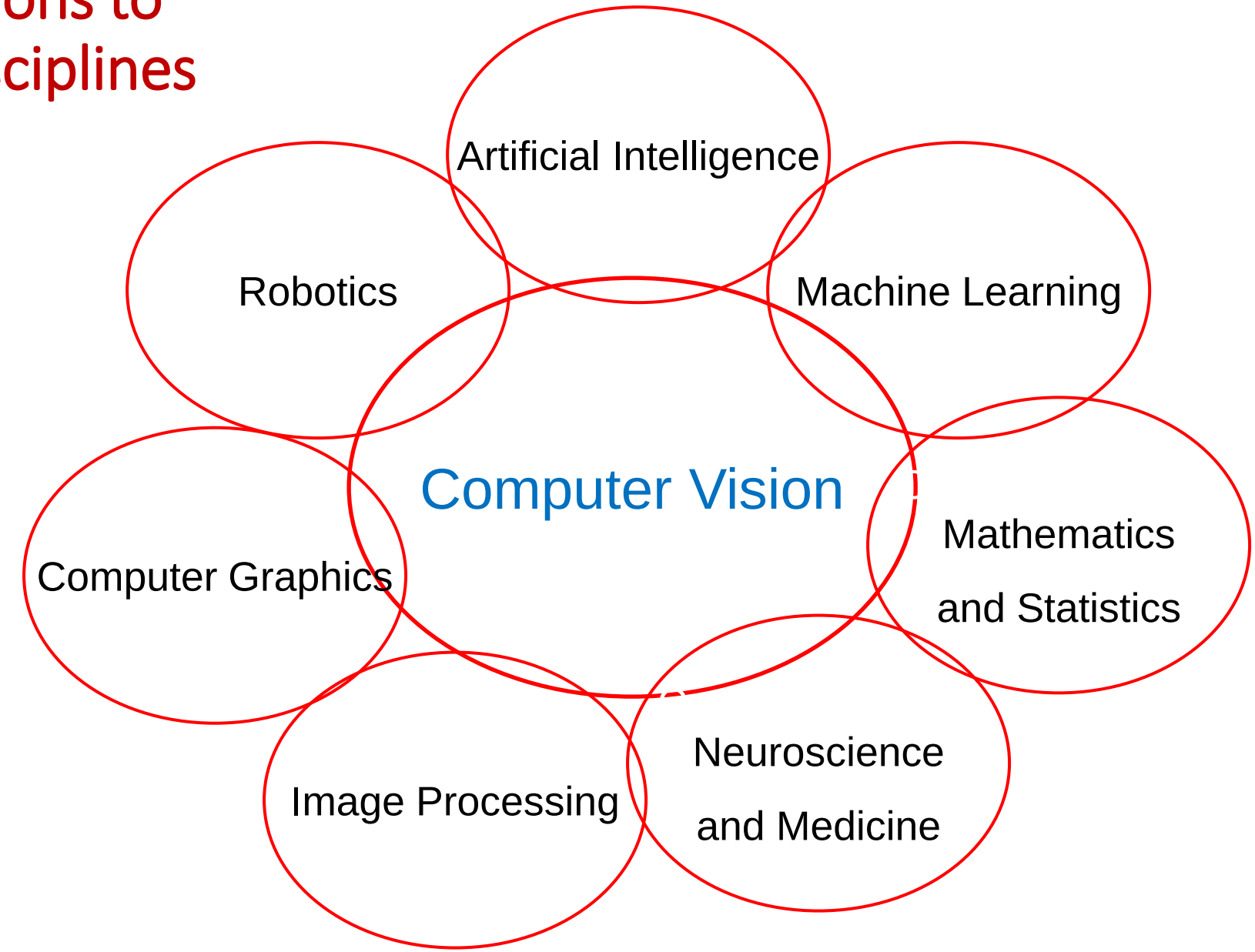




What is computer vision ?

- The bridge between pixels and “meaning”
- The inverse of graphics
- Inference: find the most probable world given visual information
- A window into mind

Connections to other disciplines



The eye is the first part of the brain

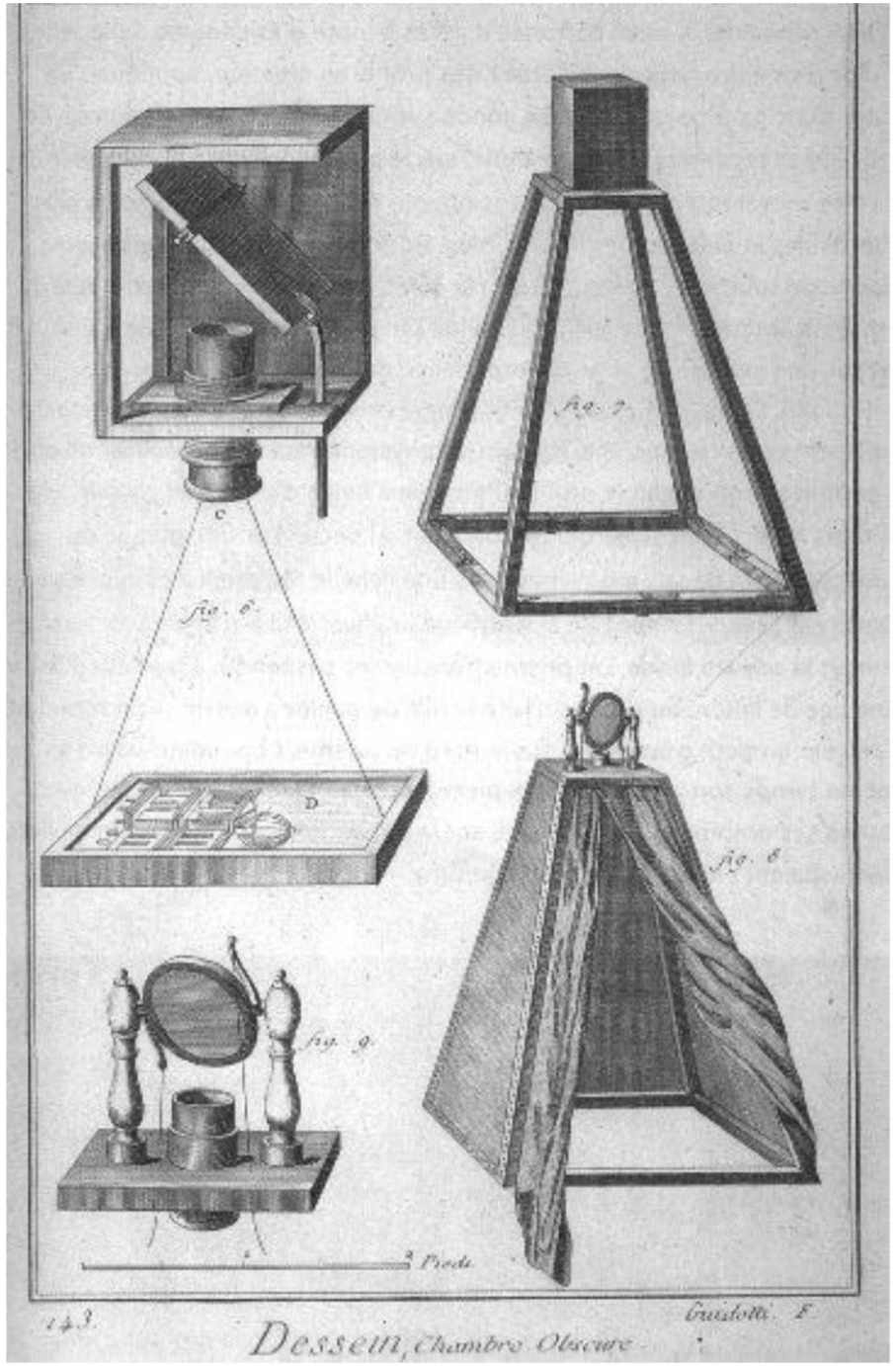
Vision is a way of thinking

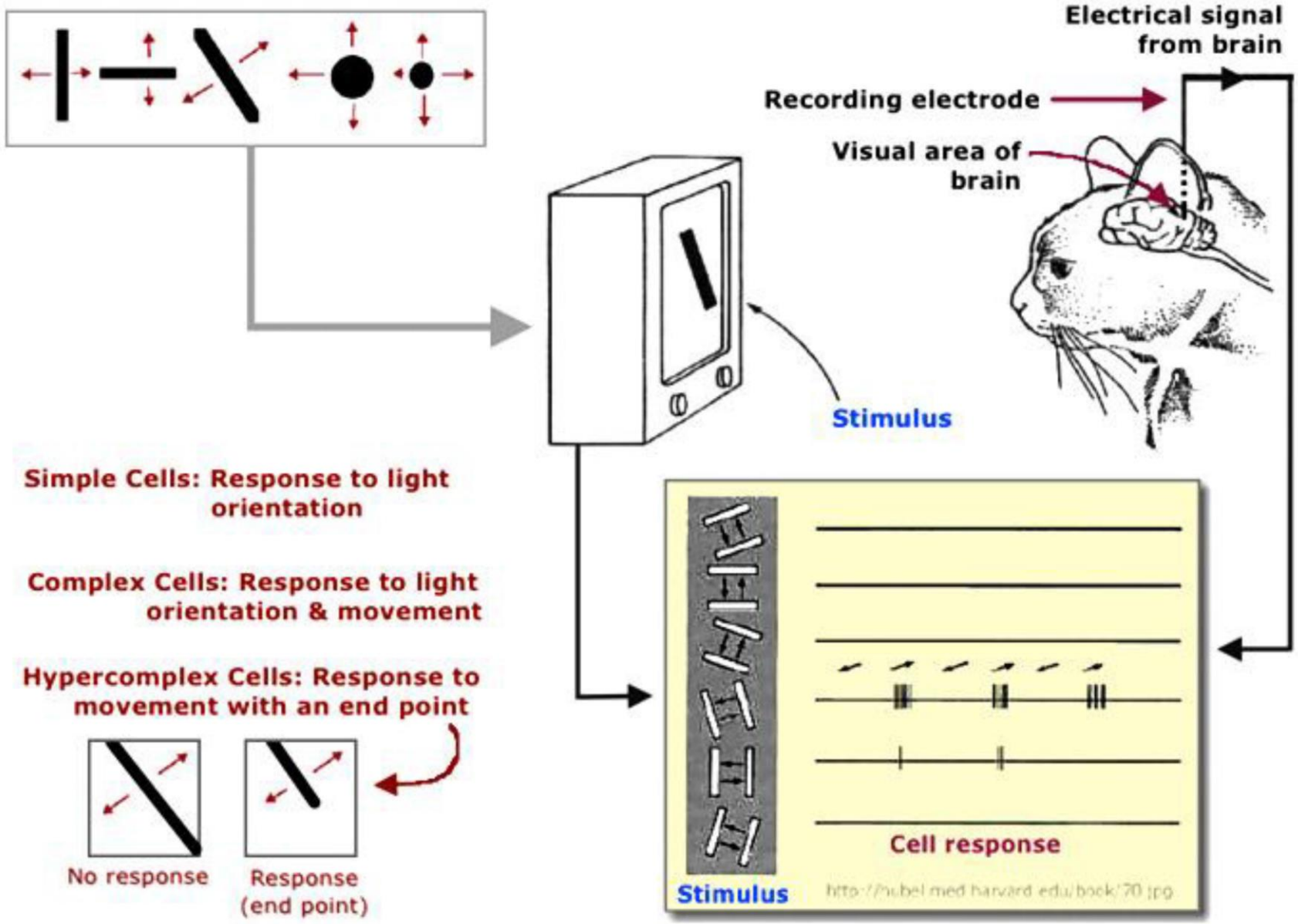
Camera Obscura

Leonardo da Vinci
16th Century, A.D.

Can we capture
the image of the world ?

What does it mean *to see* ?





Hubel & Wiesel, 1959

Visual recognition - insights from neuroscience

Hubel & Wiesel,

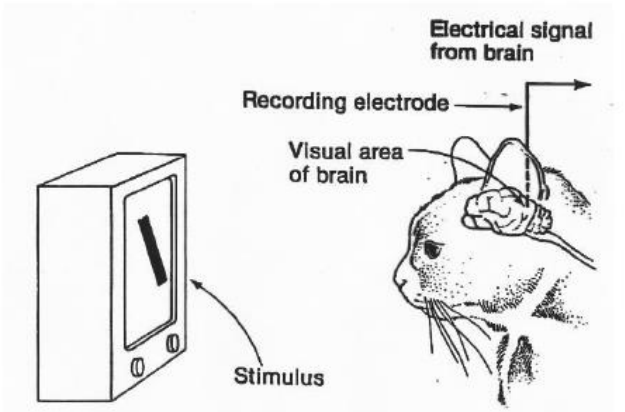
1959

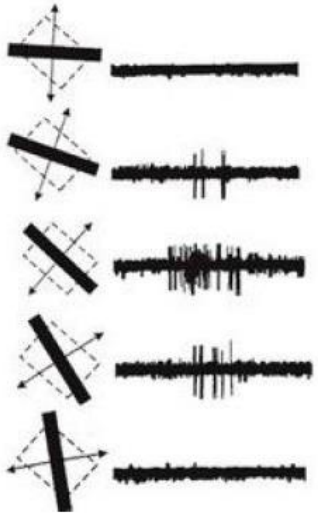
RECEPTIVE FIELDS OF SINGLE NEURONES IN THE CAT'S STRIATE CORTEX

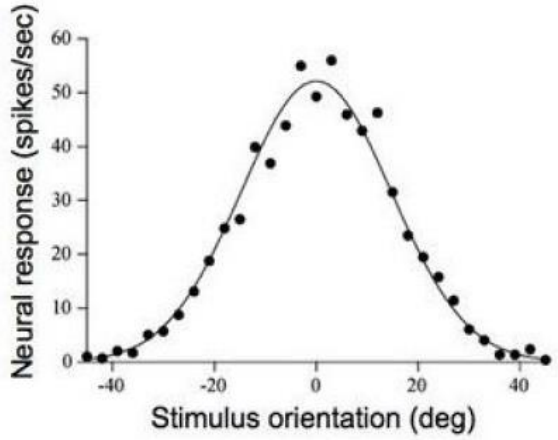
1962

RECEPTIVE FIELDS, BINOCULAR INTERACTION AND FUNCTIONAL ARCHITECTURE IN THE CAT'S VISUAL CORTEX

1968...



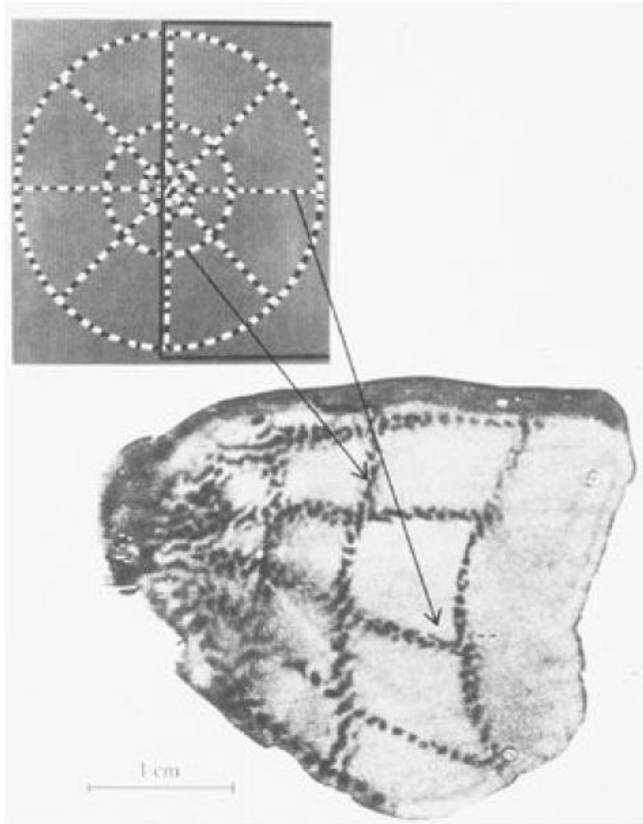




- Established a foundation for visual neurophysiology.
- Recieved Nobel Prize (1981) in Medicine for their work.
- Inspired many researchers in computer vision (e.g. David Lowe, SIFT 1999).

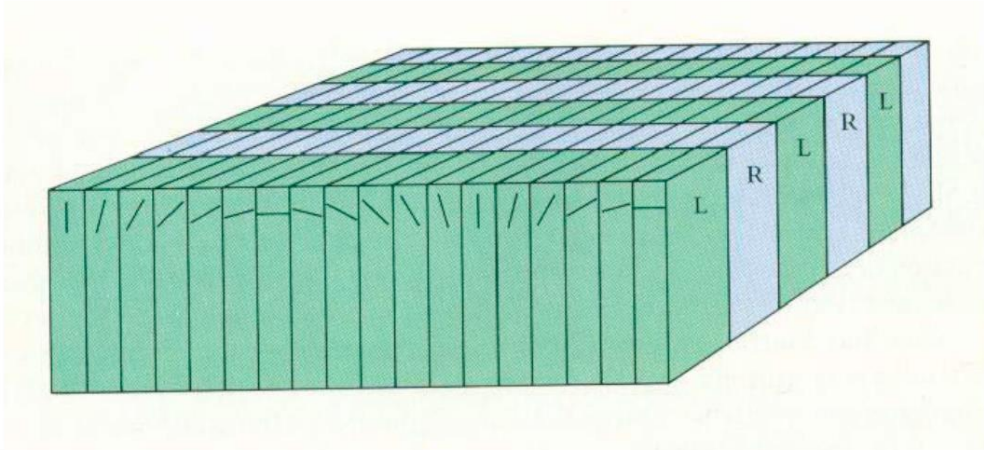
A bit of history

Topographical mapping in the cortex:
nearby cells in cortex represented
nearby regions in the visual field



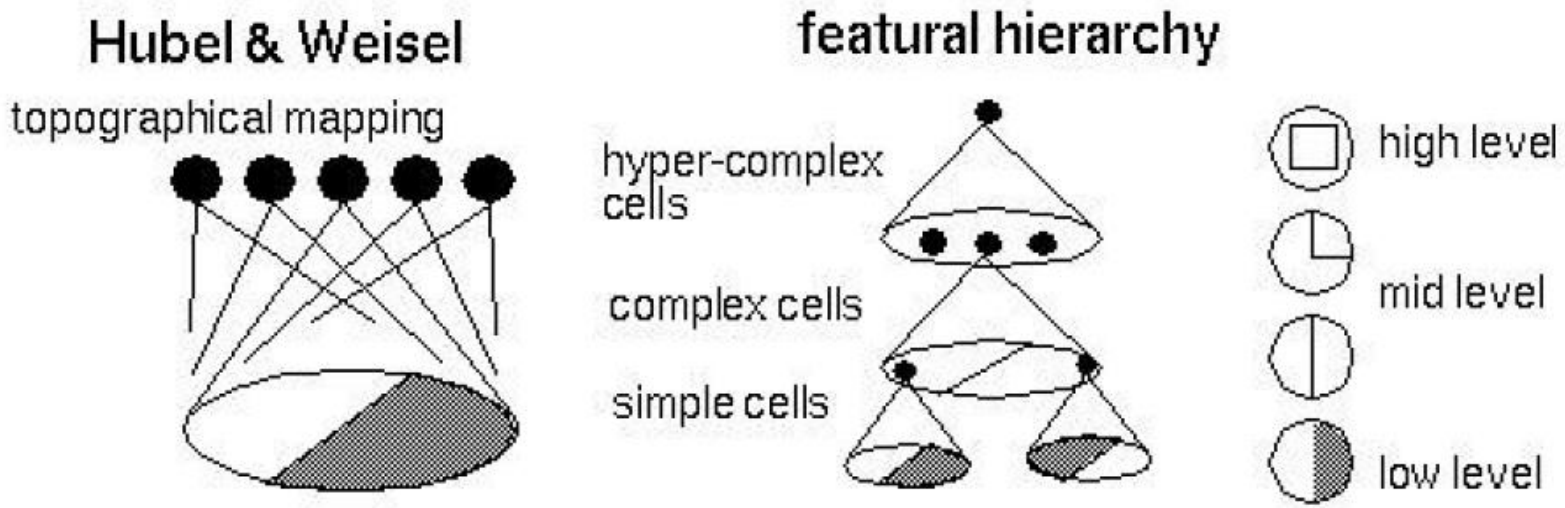
Neural Representations

- Topographic
  - Similar things represented nearby on cortex

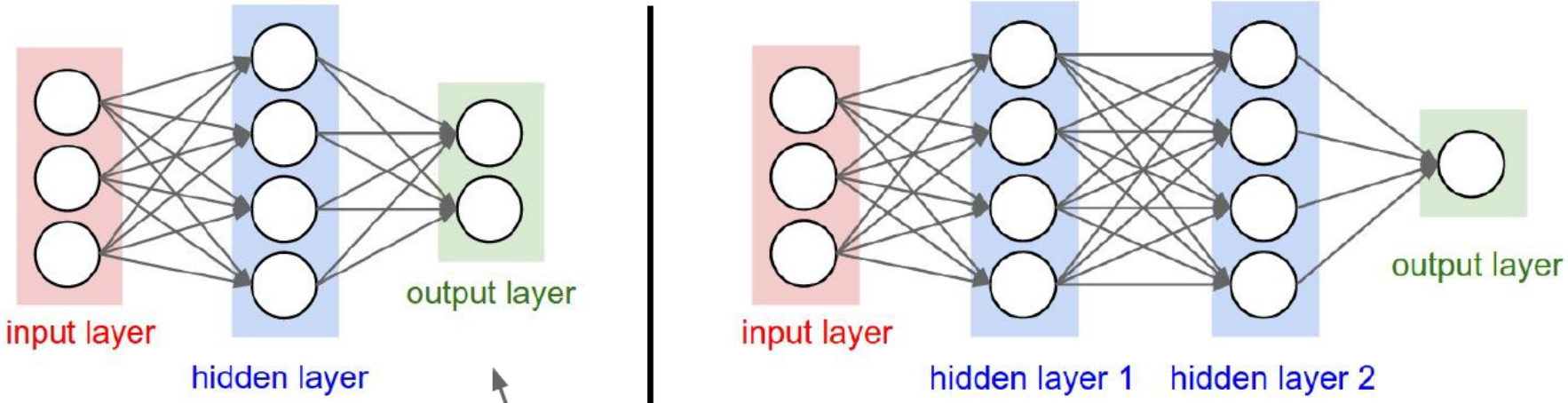


Vision is hierarchical

Hierarchical organization



Neural Networks: Architectures



"2-layer Neural Net", or
"1-hidden-layer Neural Net"

"3-layer Neural Net", or
"2-hidden-layer Neural Net"

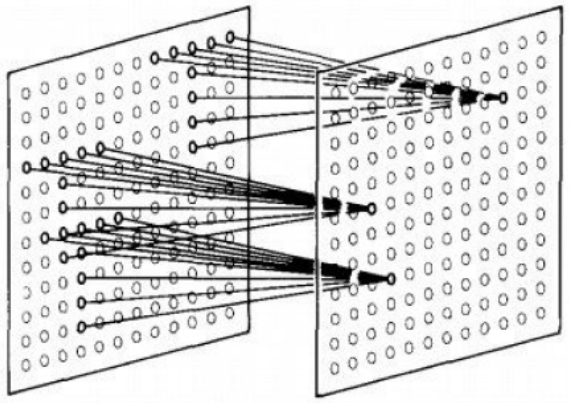
"Fully-connected" layers

Warren McCulloch and Walter Pitts (1943)
Donald Hebb (1948)
Frank Rosenblatt (1958)
Paul Werbos (1975)

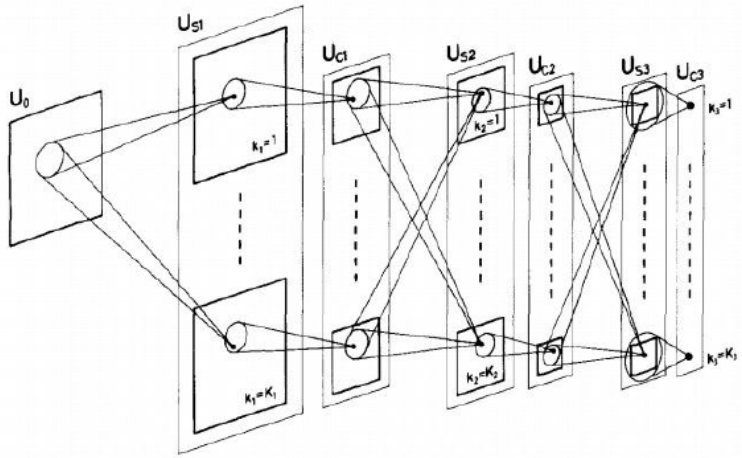
Riesenhuber, Maximilian, and Tomaso Poggio. "Hierarchical models of object recognition in cortex." *Nature neuroscience* 2.11 (1999): 1019-1025.

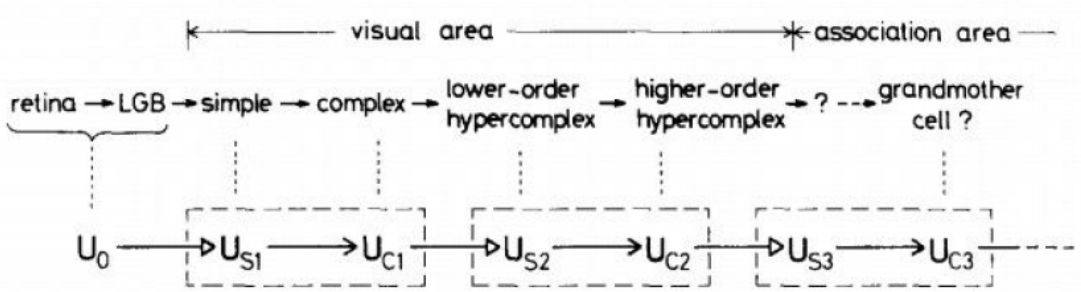
A bit of history:

Neurocognitron
[Fukushima 1980]



"sandwich" architecture (SCSCSC…)
simple cells: modifiable parameters
complex cells: perform pooling

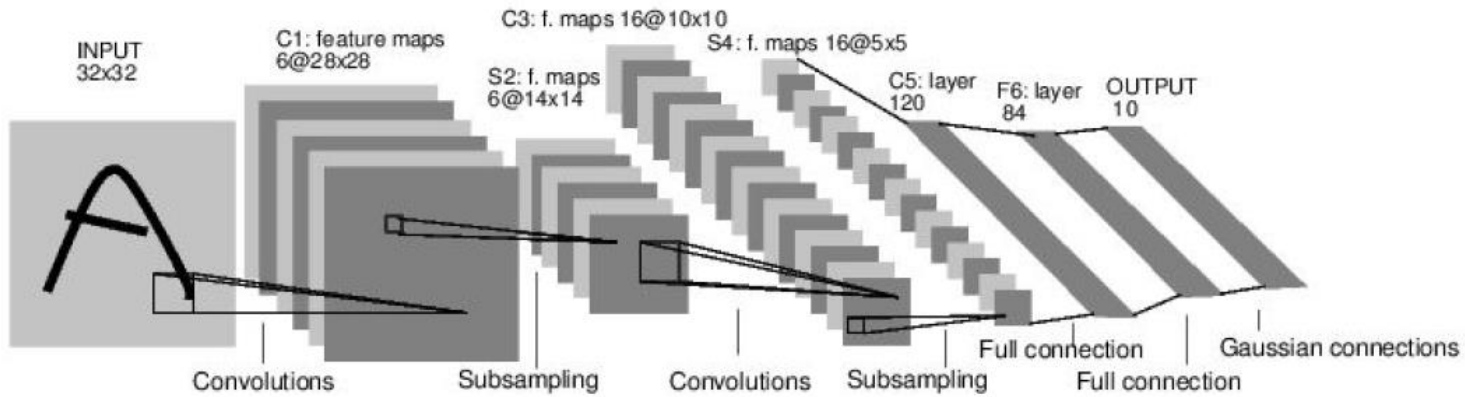




Inspired by the work of Hubel and Weisel.

Source of inspiration for Convolutional Neural Networks

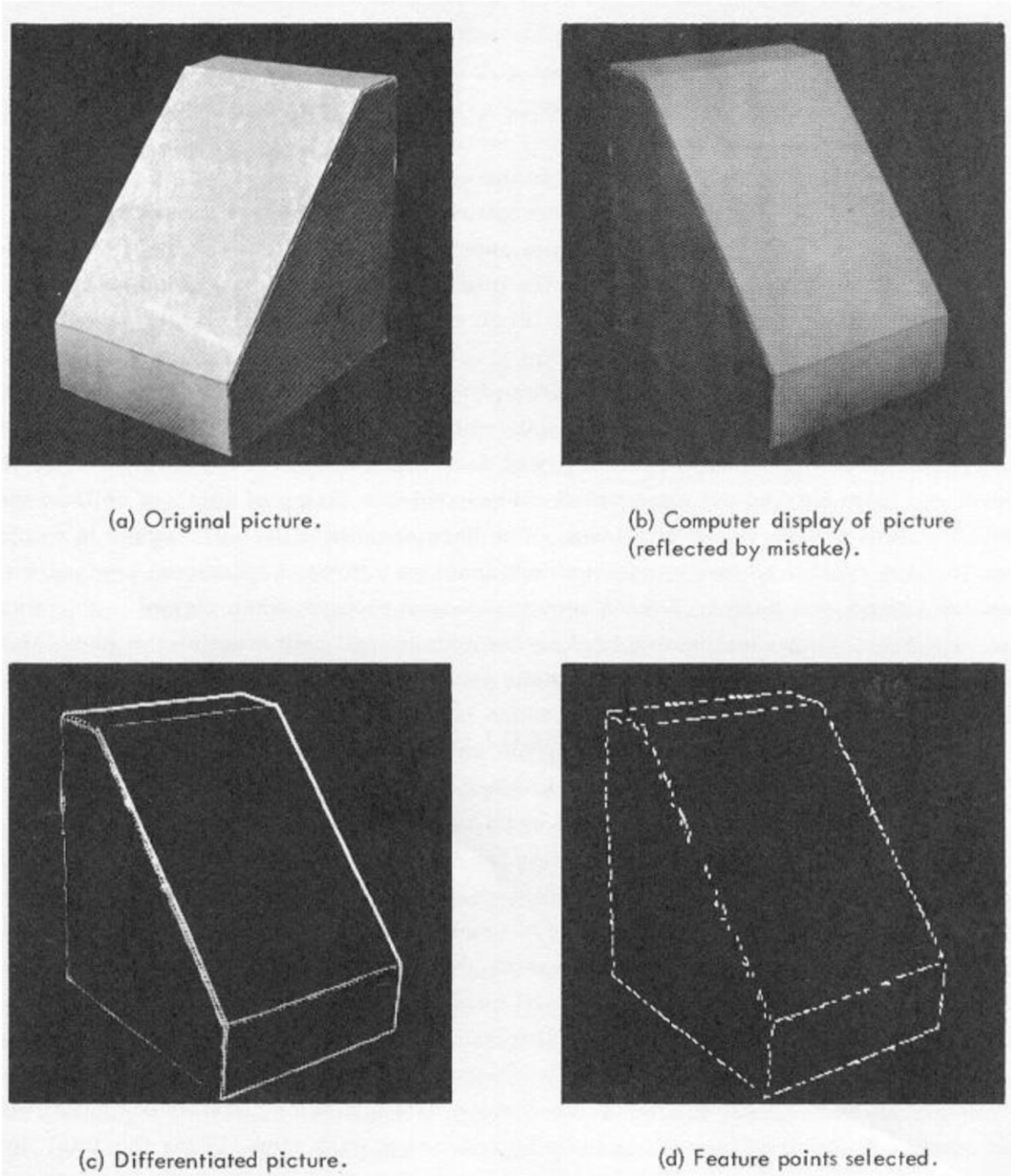
Convolutional Neural Networks



[LeNet-5, LeCun 1980]

Block world

Larry Roberts, 1963



(a) Original picture.

(b) Computer display of picture (reflected by mistake).

(c) Differentiated picture.

(d) Feature points selected.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

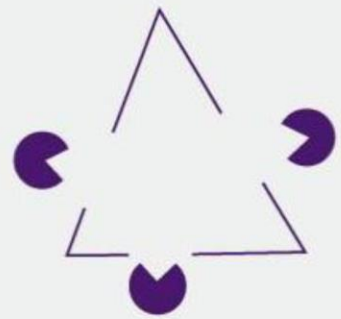
THE SUMMER VISION PROJECT

Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

Copyrighted Material

VISION



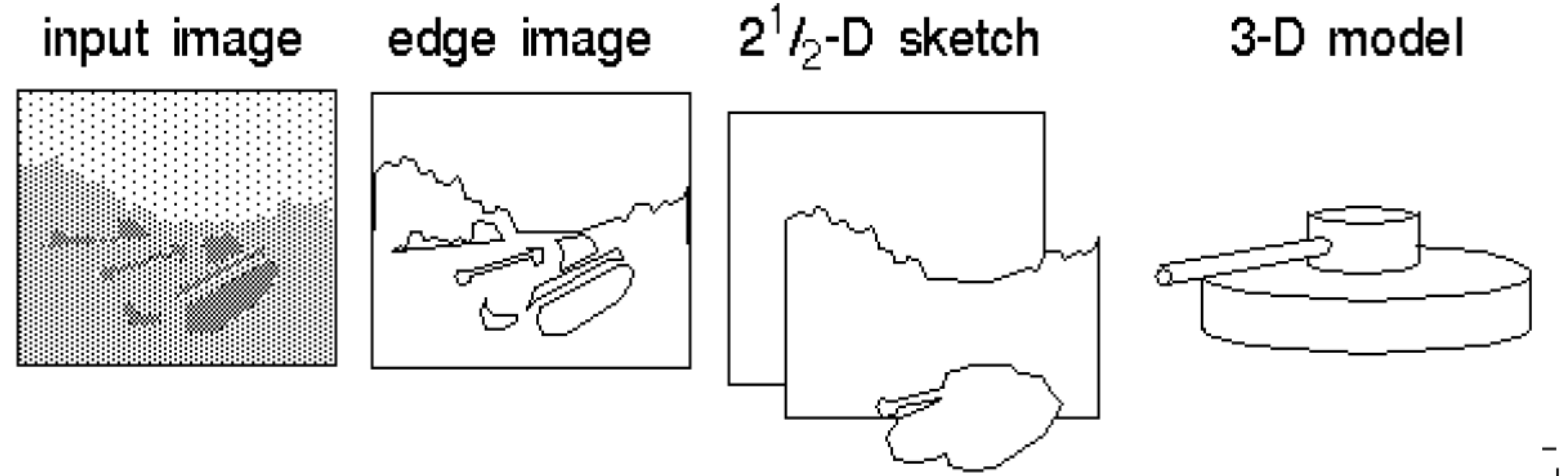
David Marr

FOREWORD BY
Shimon Ullman

AFTERWORD BY
Tomaso Poggio

Copyrighted Material

David Marr, 1970s

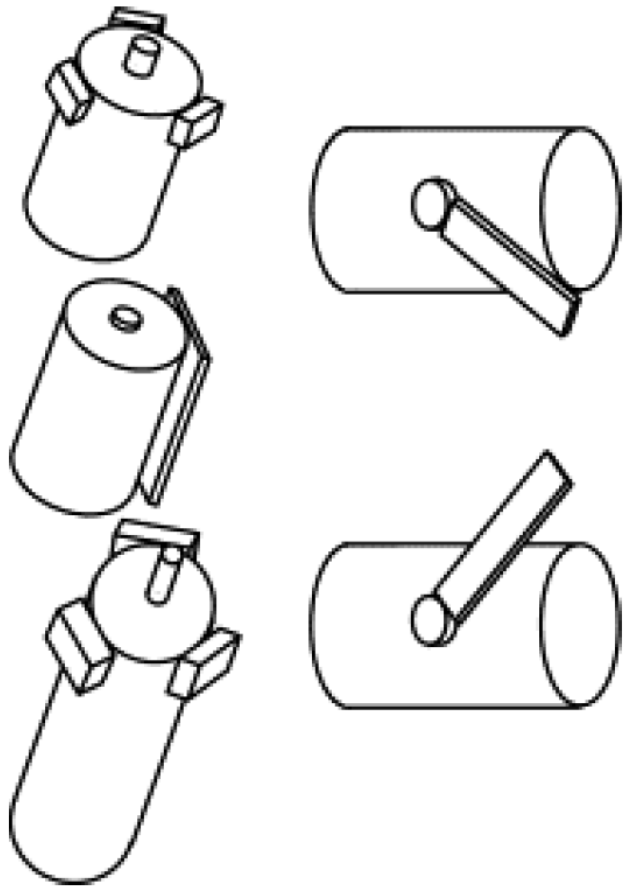


Input Image	Primal Sketch	2 1/2-D Sketch	3-D Model Representation
Perceived intensities	Zero crossings, blobs,edges, bars, ends, virtual lines, groups, curves boundaries.	Local surface orientation and discontinuities in depth and in surface orientation	3-D models hierarchically organised in terms of surface and volumetric primitives

Stages of Visual Representation, David Marr,

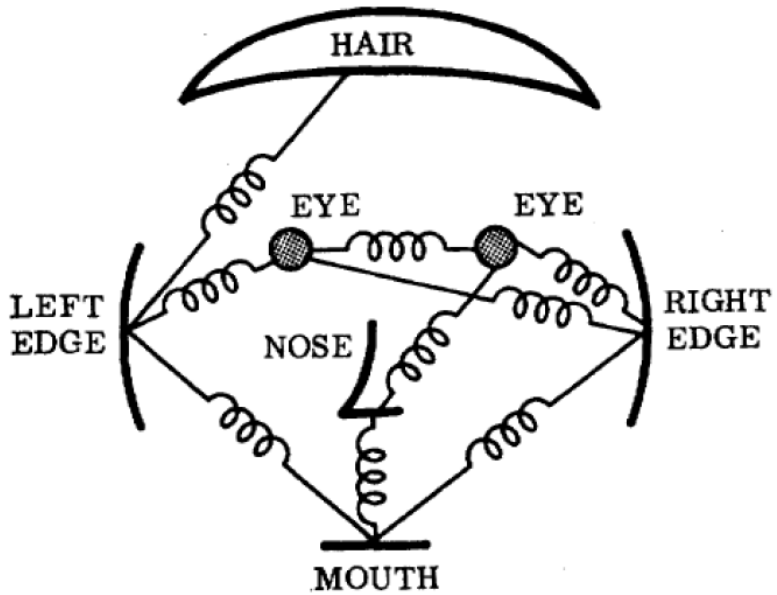
- Generalized Cylinder

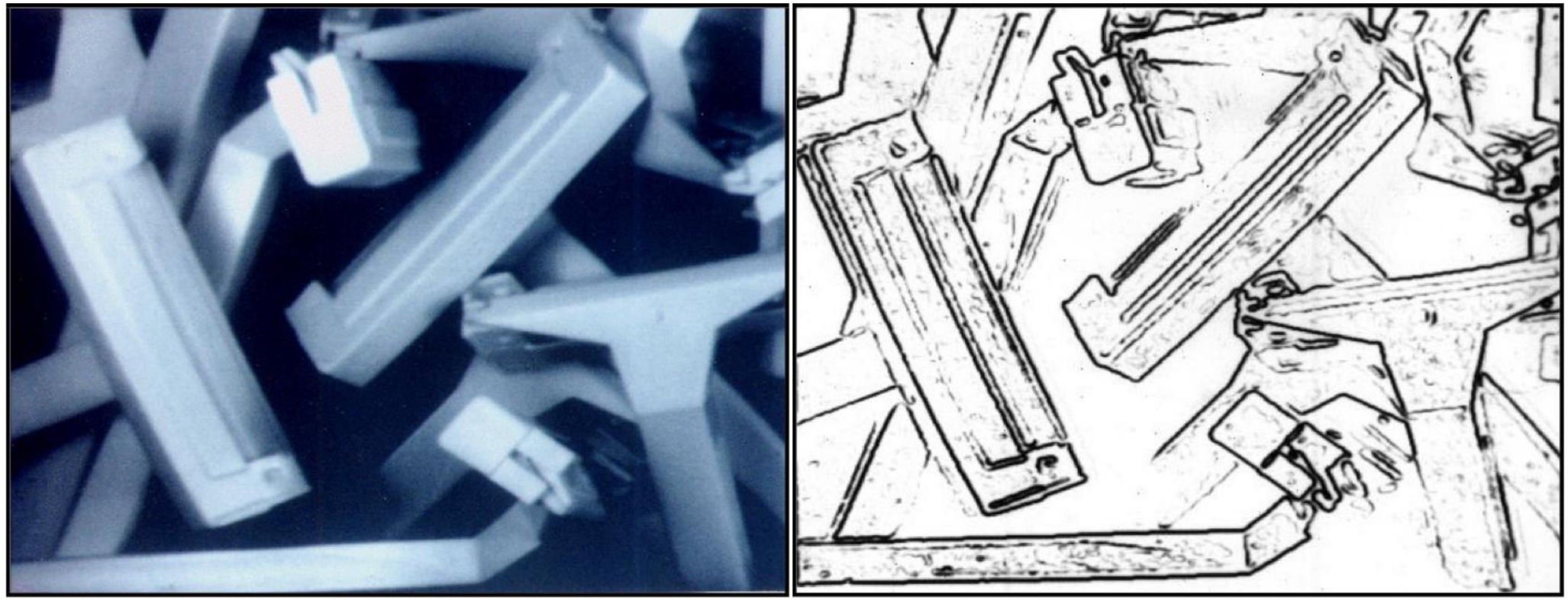
Brooks & Binford, 1979



- Pictorial Structure

Fischler and Elschlager, 1973



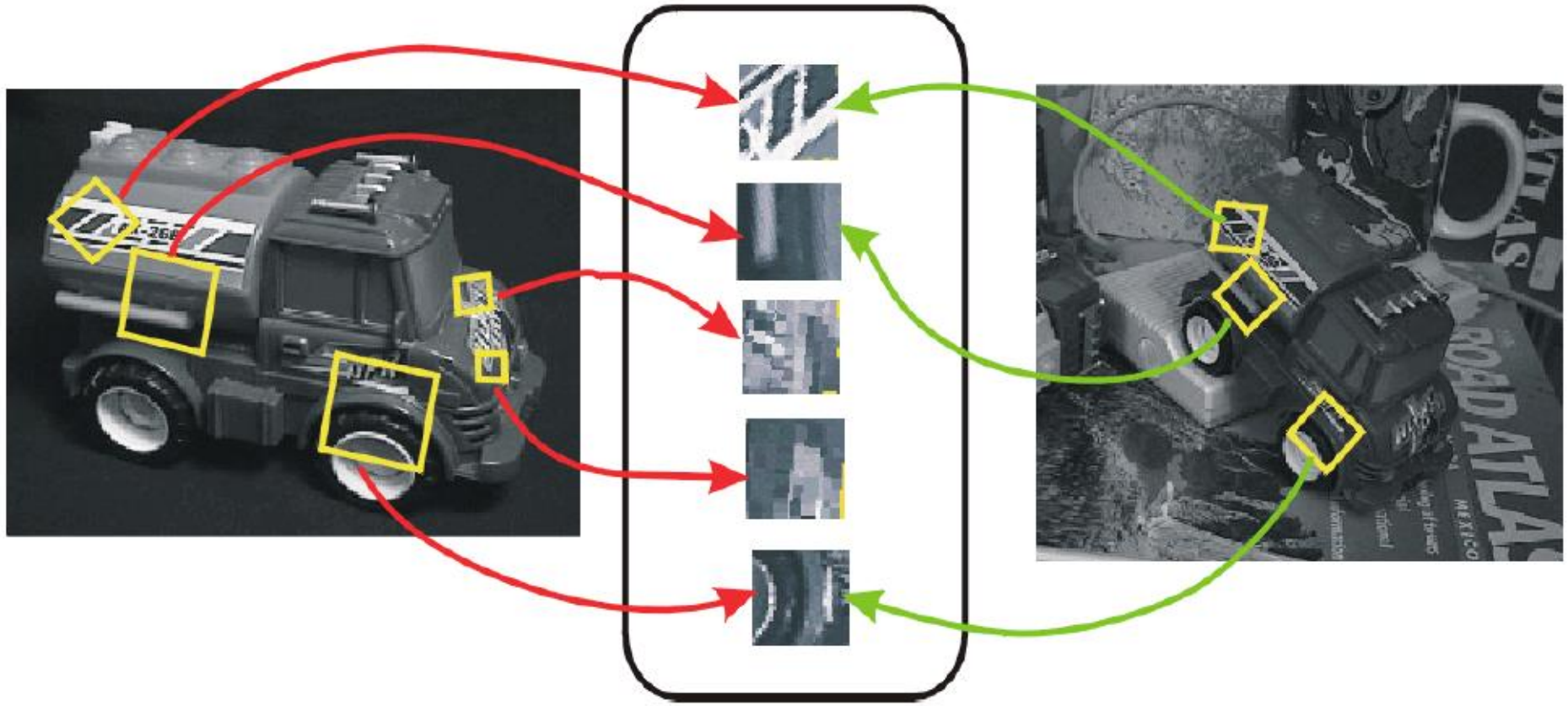


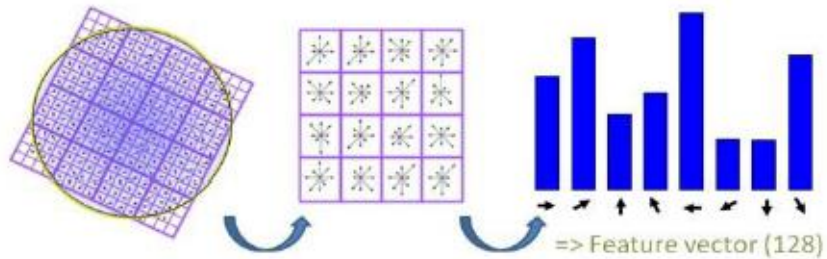
David Lowe, 1987


Normalized Cut
(Shi & Malik, 1997)

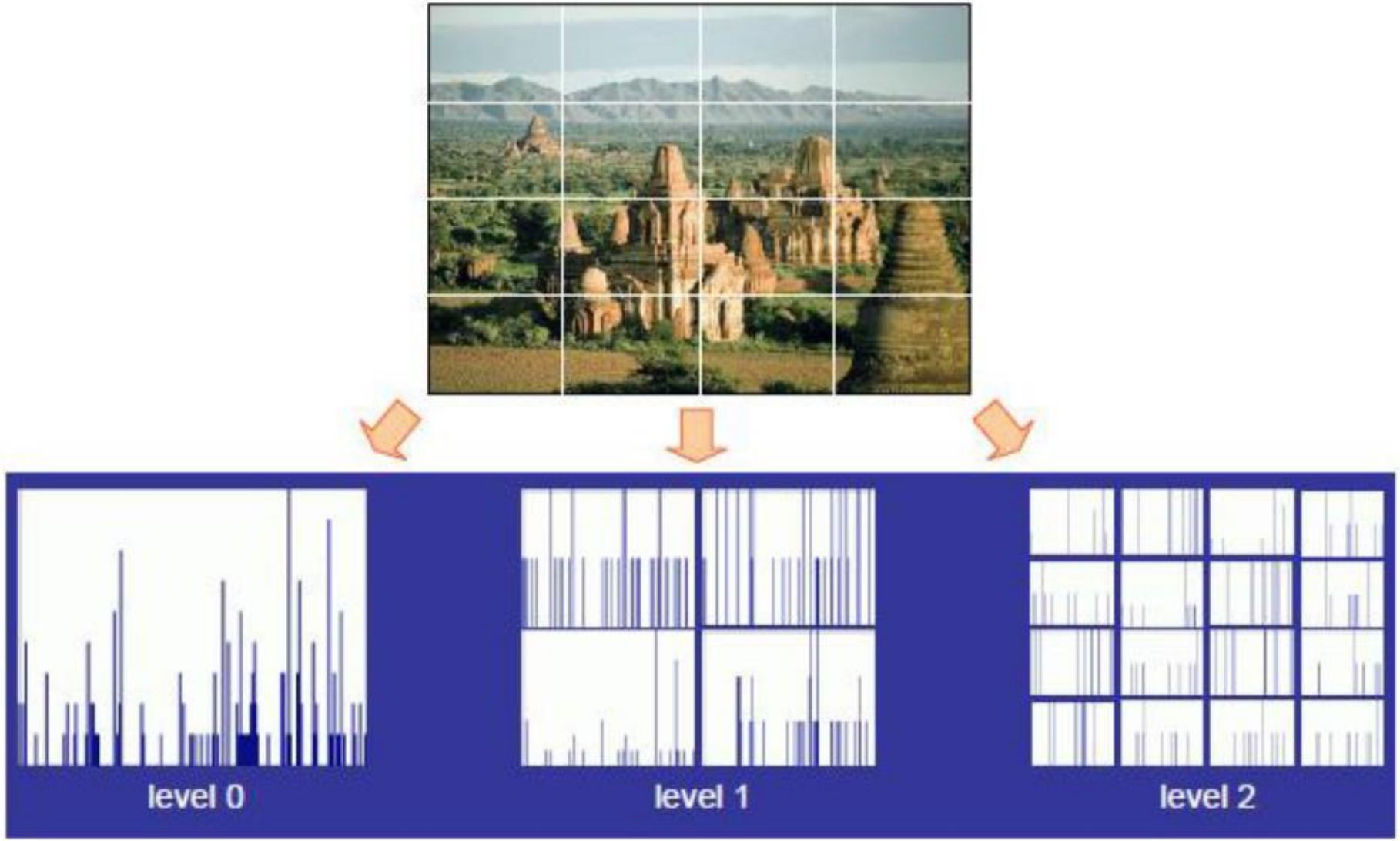


Face Detection, Viola & Jones, 2001

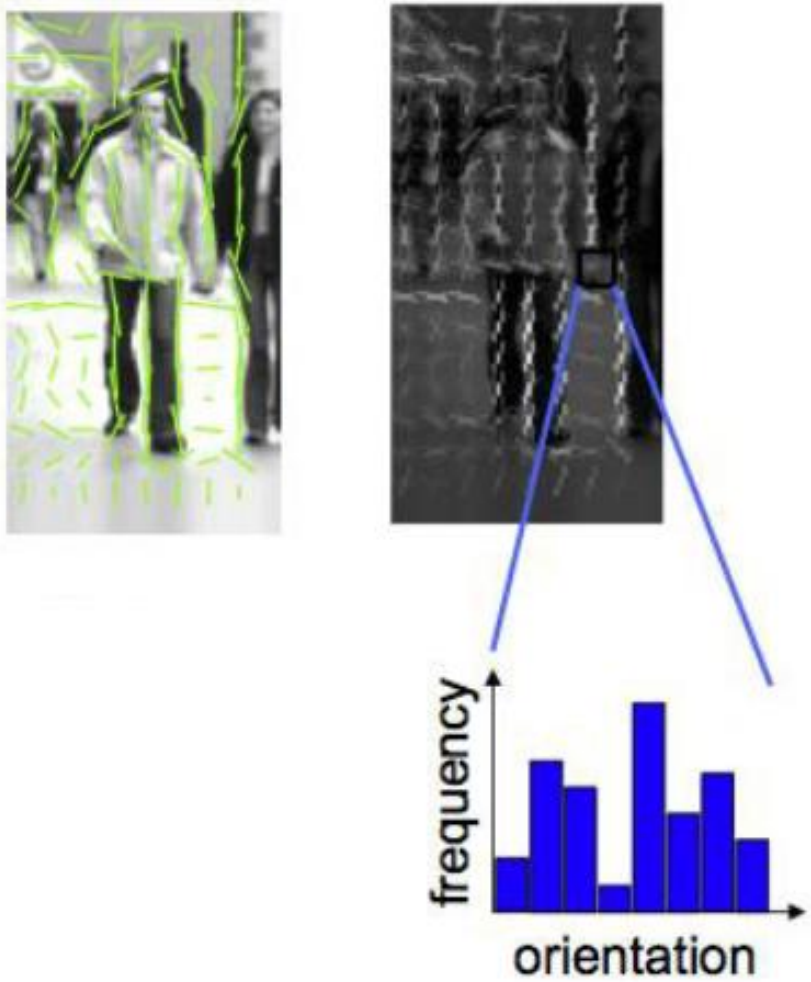




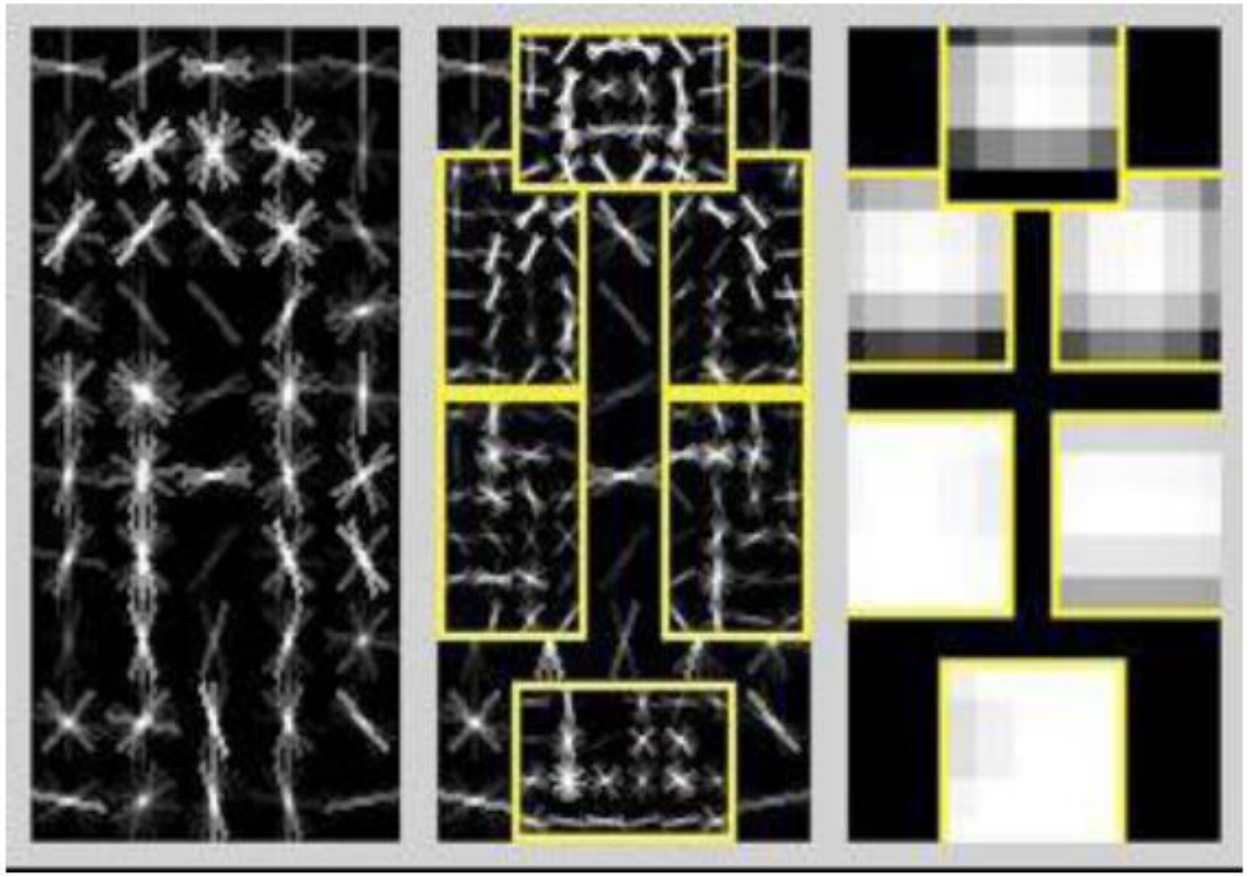
"SIFT" & Object Recognition, David Lowe, 1999



Spatial Pyramid Matching, Lazebnik, Schmid & Ponce, 2006



Histogram of Gradients (HoG)
Dalal & Triggs, 2005

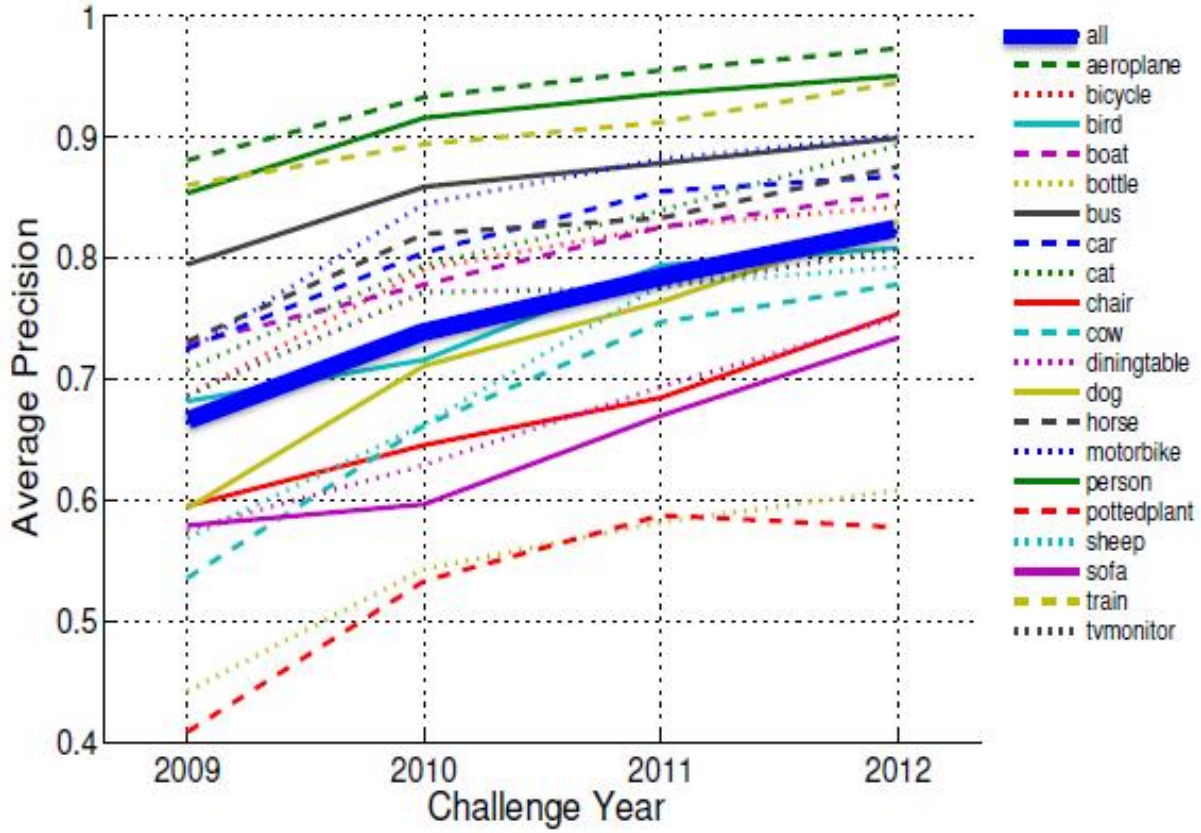


Deformable Part Model
Felzenswalb, McAllester, Ramanan, 2009

PASCAL Visual Object Challenge (20 object categories)

[Everingham et al. 2006-2012]





IMAGENET Large Scale Visual Recognition Challenge

Steel drum

The Image Classification Challenge:

1,000 object classes

1,431,167 images



Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

✔

Output:
Scale
T-shirt
Giant panda
Drumstick
Mud turtle

✗

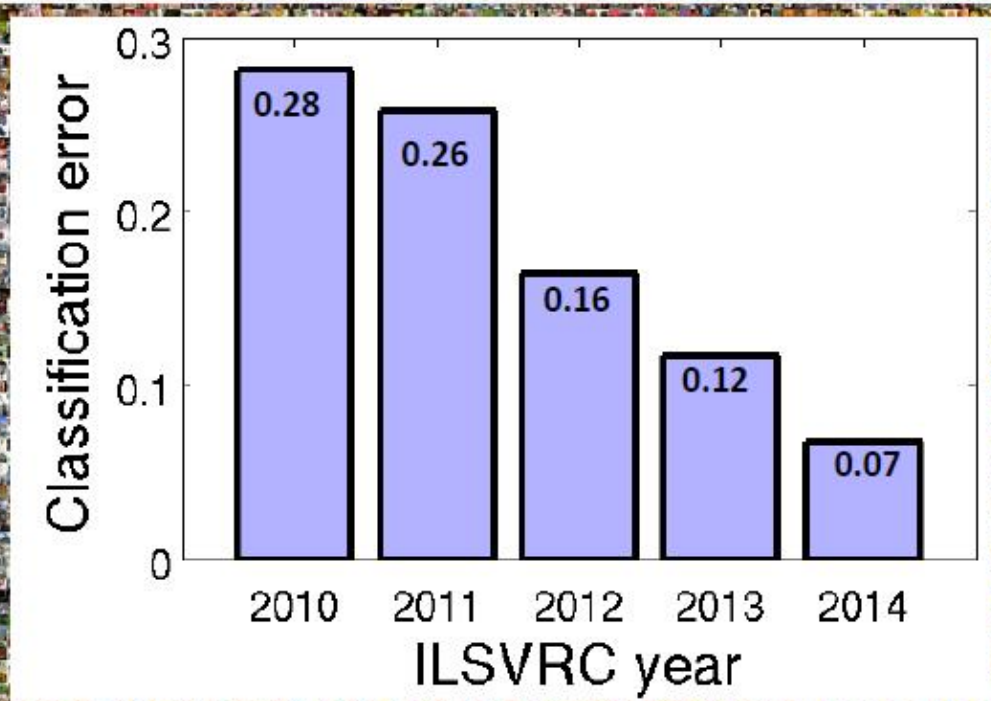
Russakovsky et al. arXiv, 2014

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:

1,000 object classes

1,431,167 images

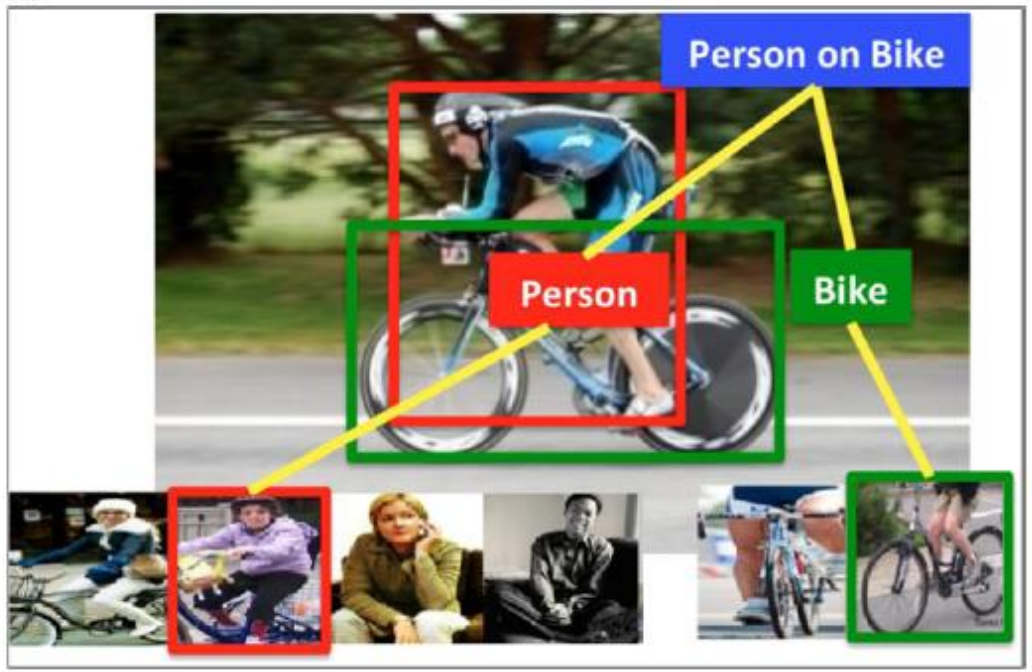


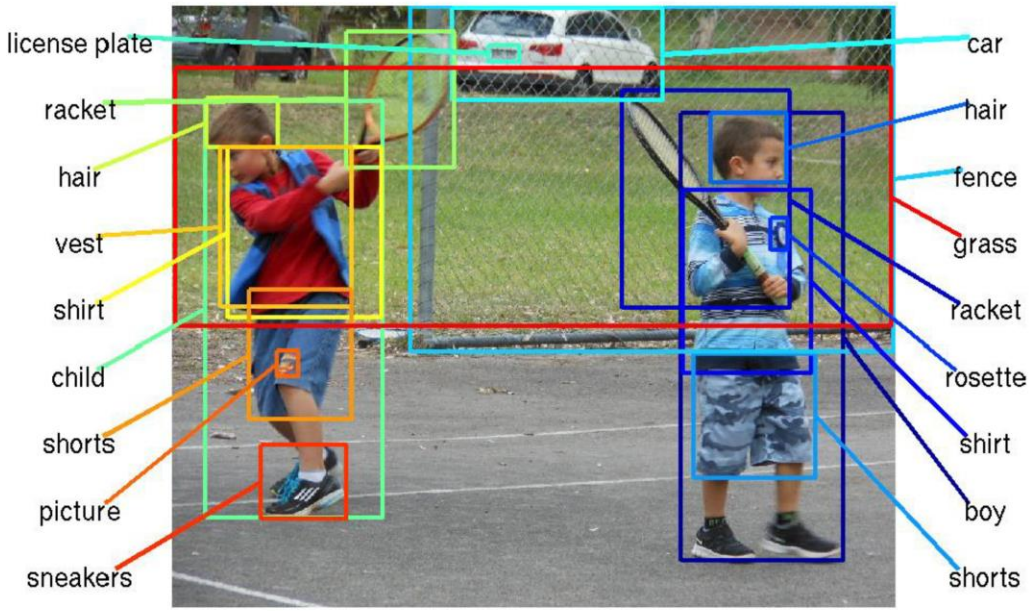
Russakovsky et al. arXiv, 2014

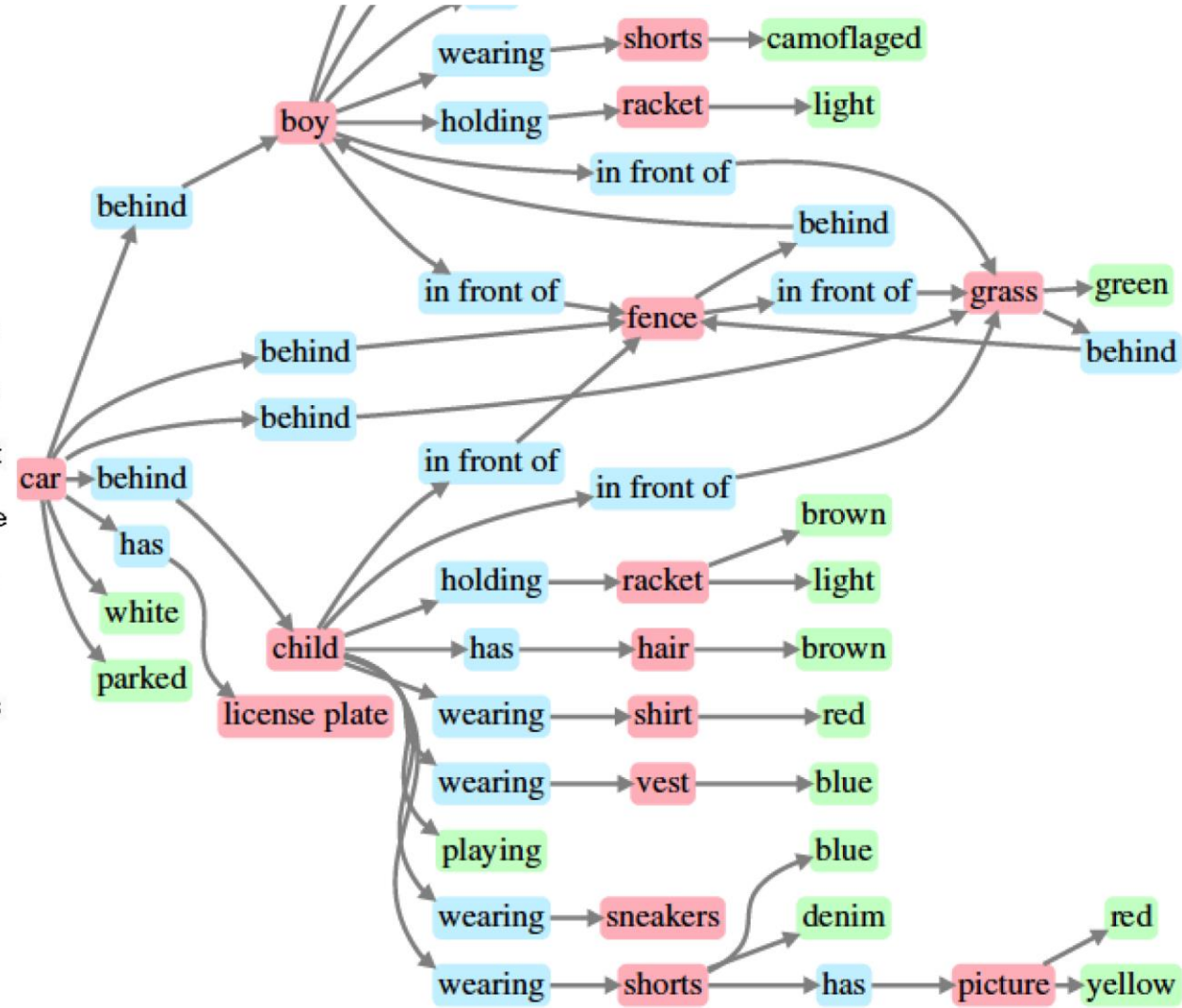


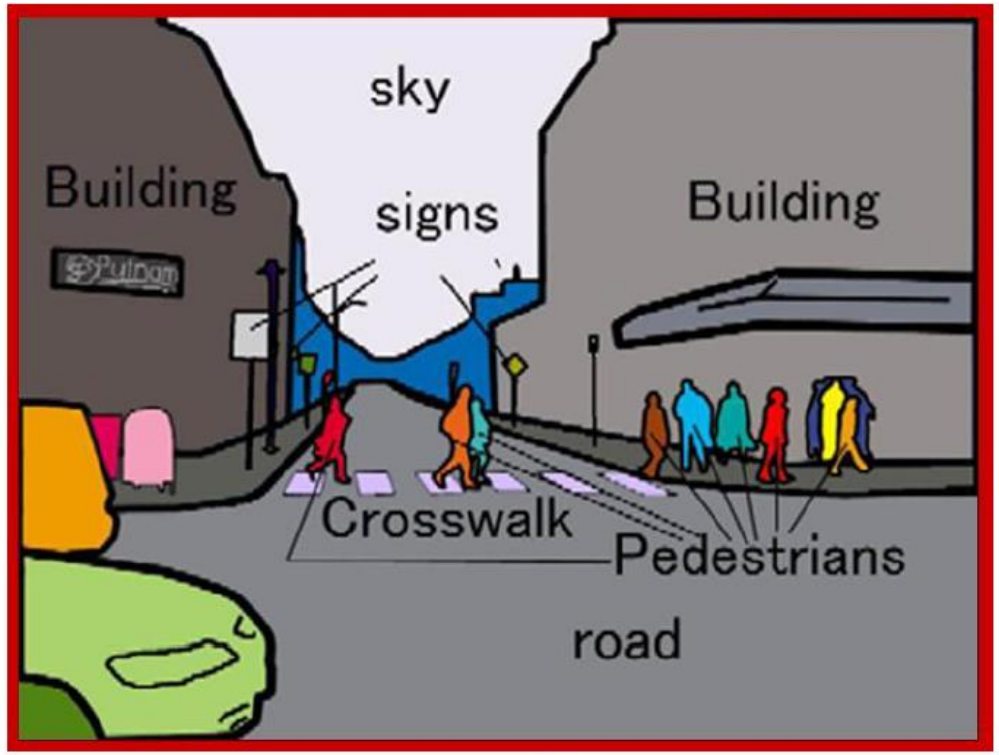
- Object detection
- Action classification
- Image captioning
- …

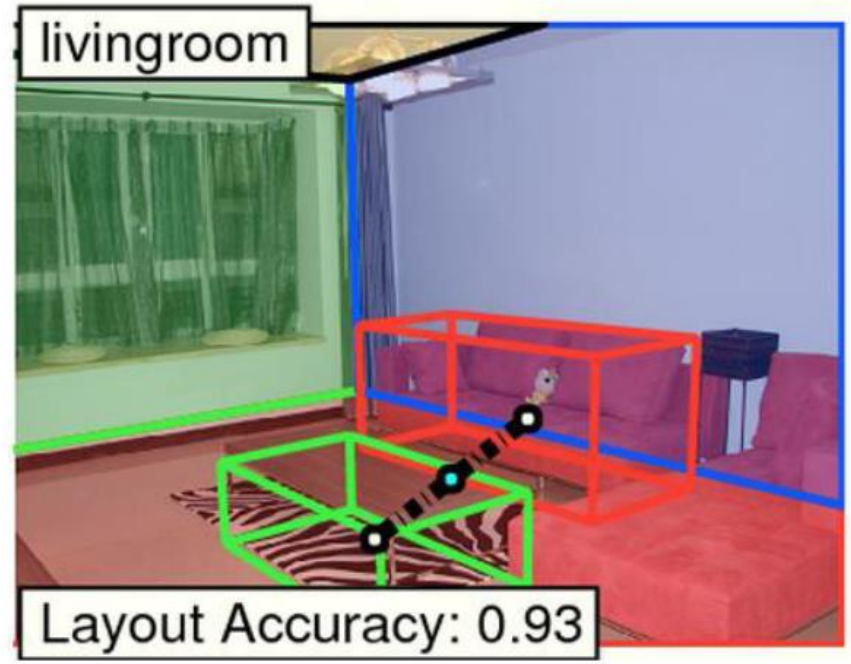


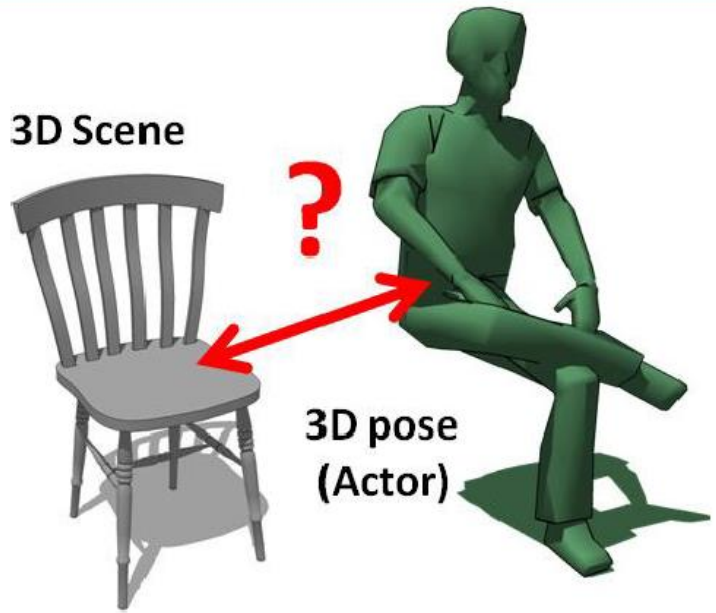


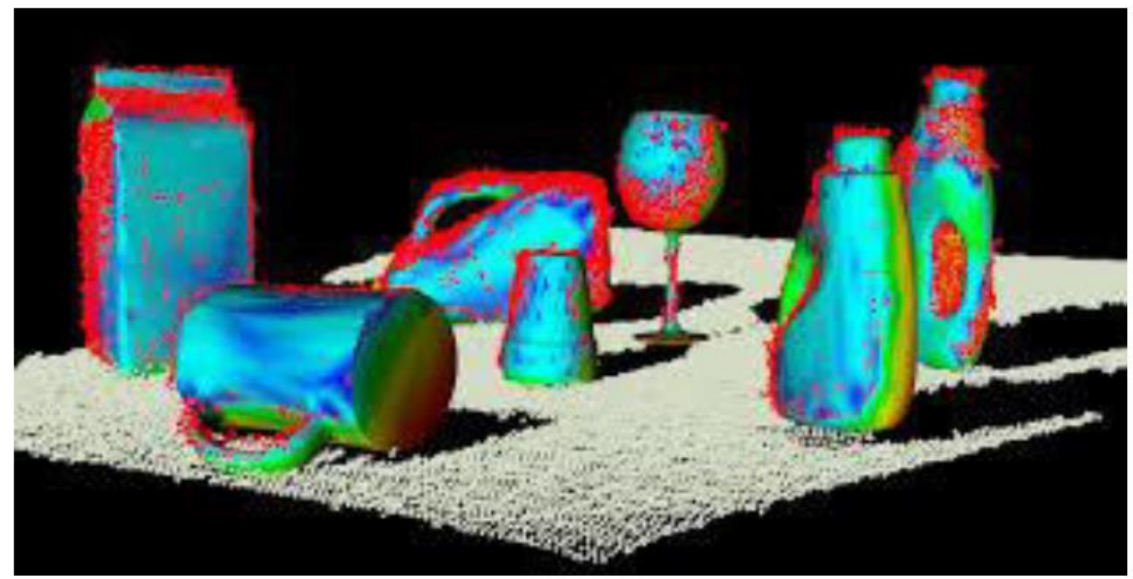

license plate
racket
hair
vest
shirt
child
shorts
picture
sneakers
car
hair
fence
grass
racket
rosette
shirt
boy
shorts


boy
wearing
shorts
camoflaged
holding
racket
light
in front of
behind
behind
in front of
fence
in front of
grass
green
behind
behind
behind
car
behind
in front of
in front of
has
white
parked
child
holding
racket
brown
light
has
hair
brown
license plate
wearing
shirt
red
wearing
vest
blue
playing
blue
wearing
sneakers
denim
wearing
shorts
has
picture
red
yellow


sky
Building
signs
Building
Crosswalk
Pedestrians
road


livingroom
Layout Accuracy: 0.93


3D Scene
?
3D pose
(Actor)



IMAGENET Large Scale Visual Recognition Challenge

Year 2010

NEC-UIUC



Dense grid descriptor: HOG, LBP

Coding: local coordinate, super-vector

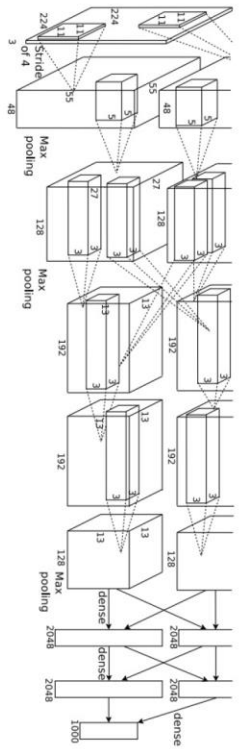
Pooling, SPM

Linear SVM

[Lin CVPR 2011]

Year 2012

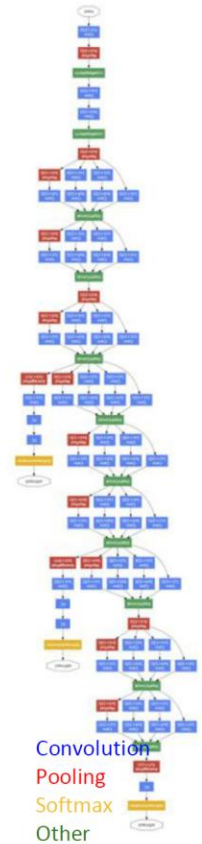
SuperVision

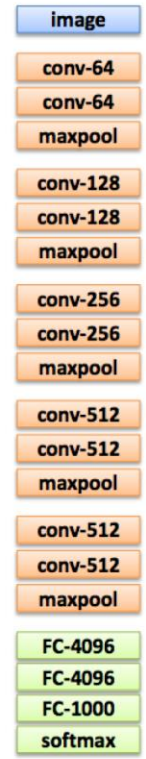


[Krizhevsky NIPS 2012]

Year 2014

GoogLeNet VGG

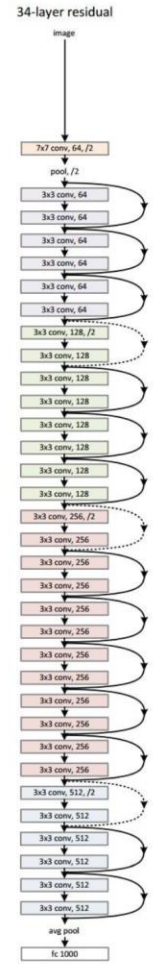




[Szegedy arxiv 2014] [Simonyan arxiv 2014]

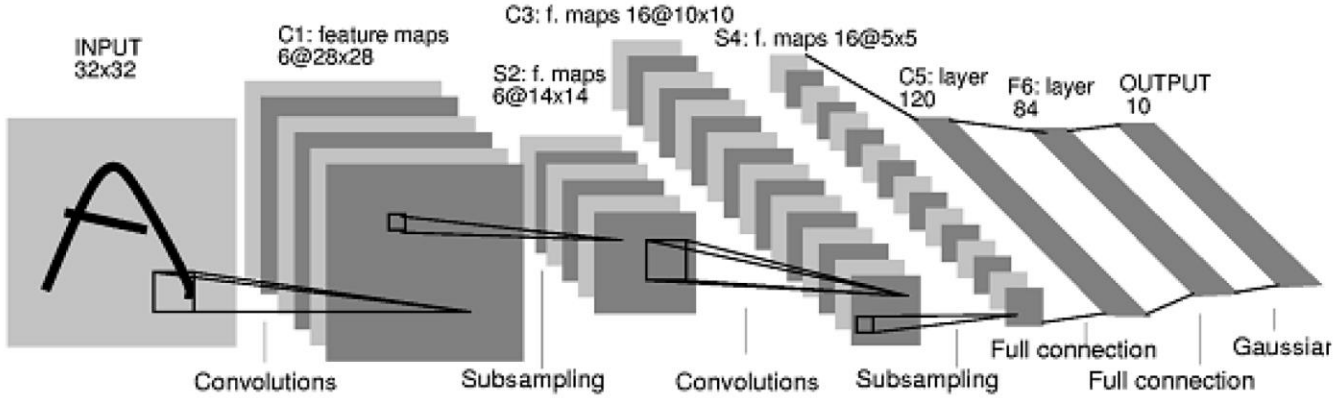
Year 2015

MSRA



1998

LeCun et al.



of transistors



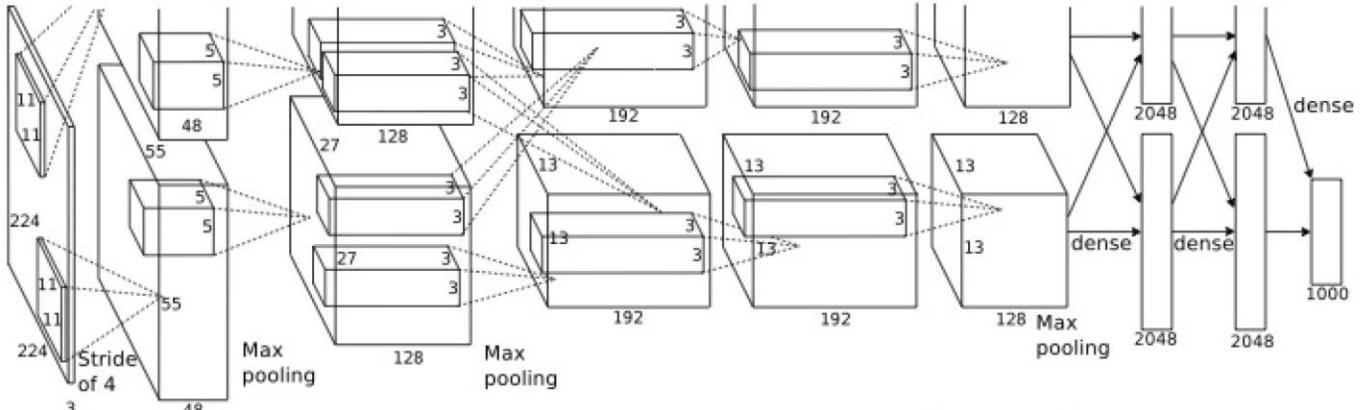
10^6

of pixels used in training

10^7 NIST

2012

Krizhevsky et al.



of transistors GPUs



10^9



of pixels used in training

10^{14} IMAGENET

Trends in computer vision and natural language processing today

Vision Transformers as Masked Autoencoders (MAE)

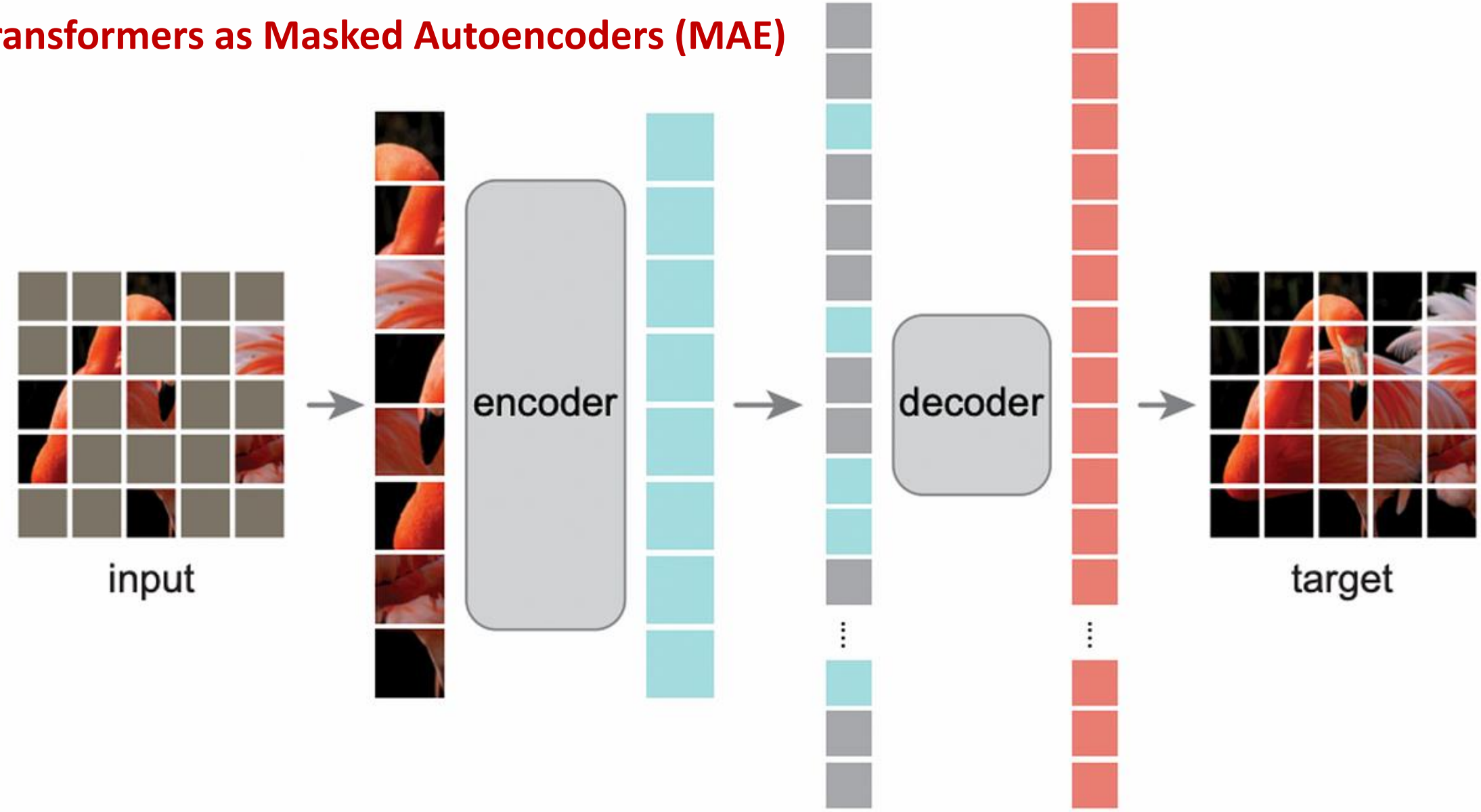


Image Classification: a core task in Computer Vision



(assume given set of discrete labels)
{dog, cat, truck, plane, ...}

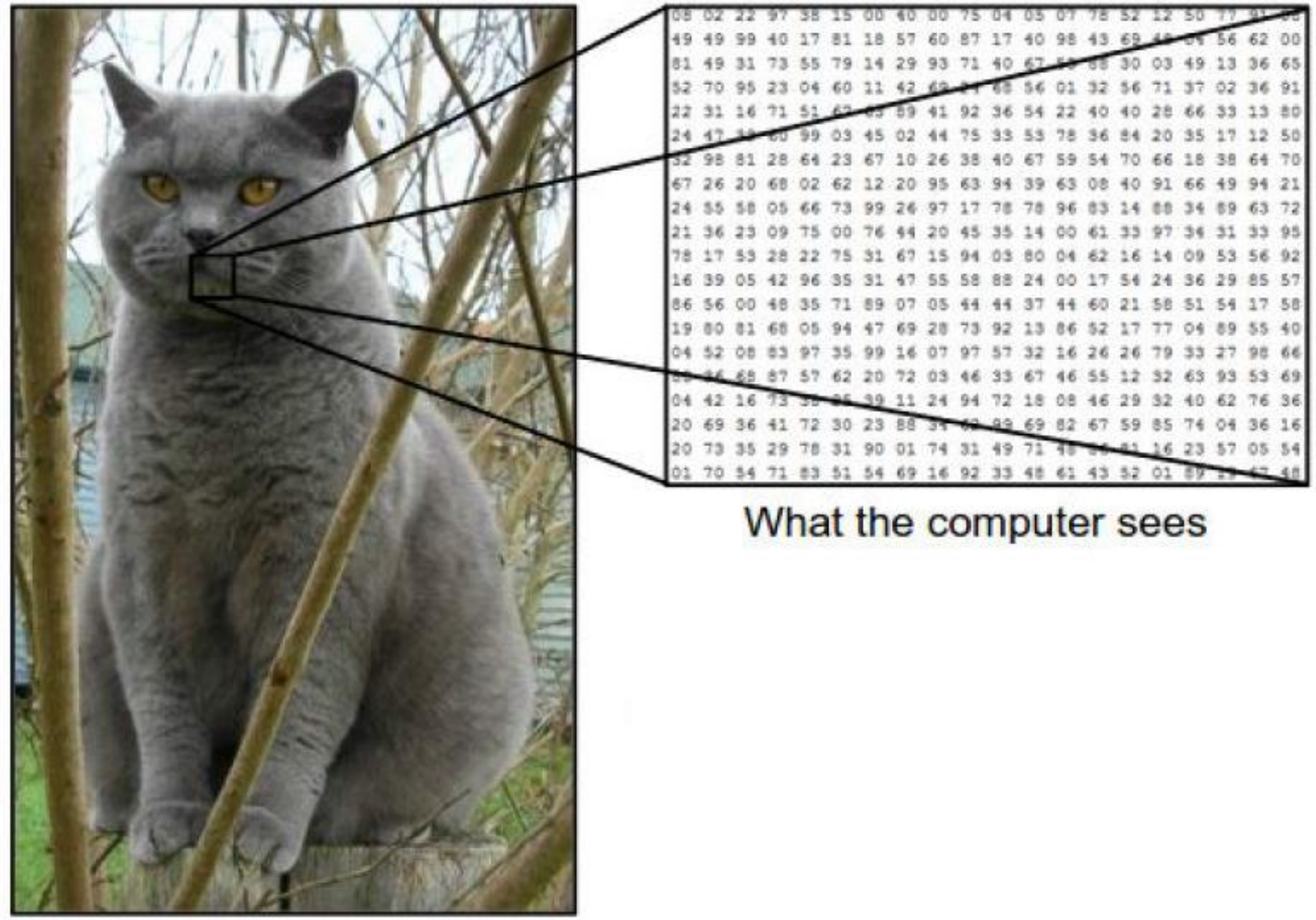
cat

The problem:
semantic gap

Images are represented as 3D arrays of numbers, with integers between [0, 255].

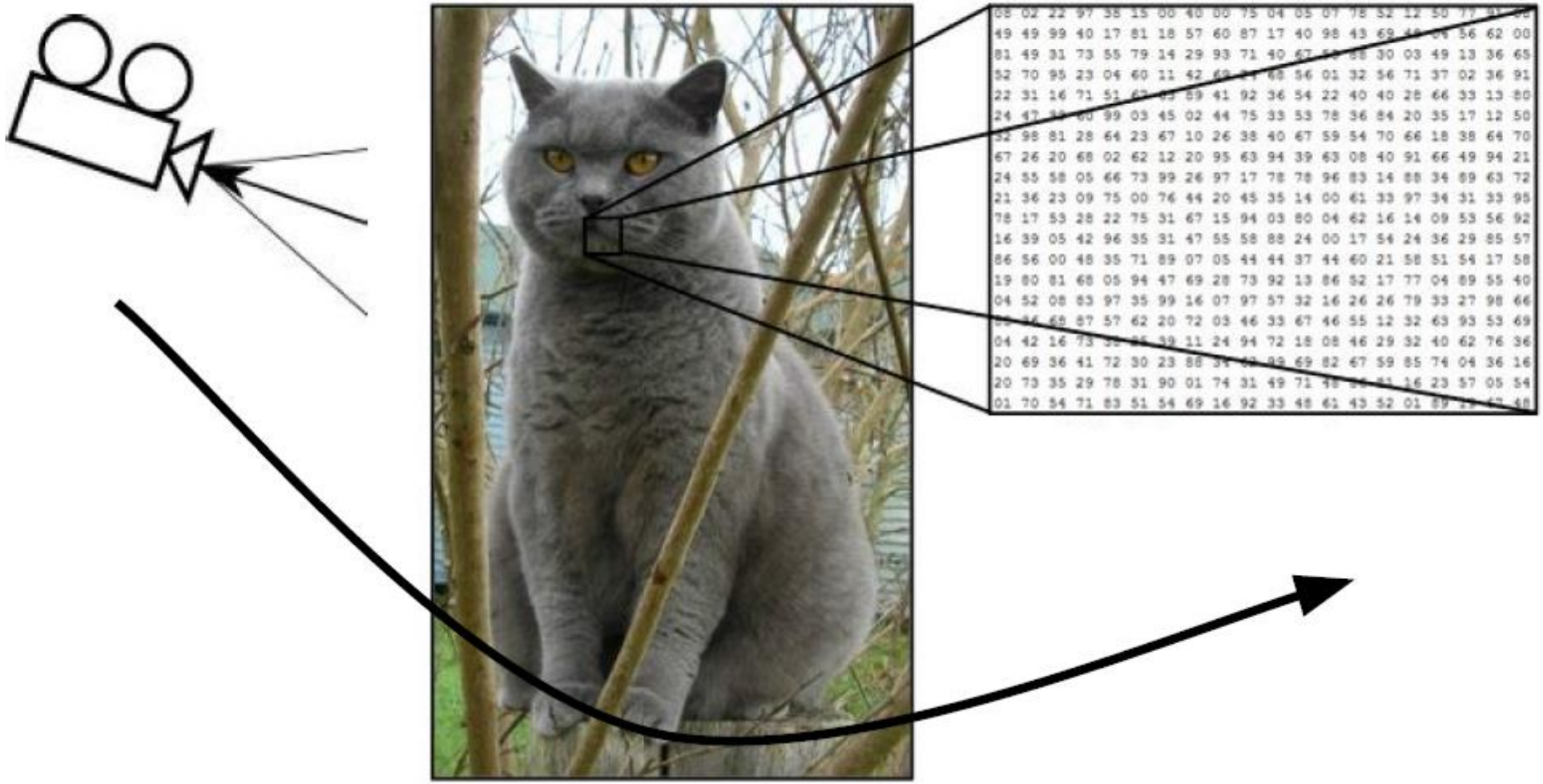
E.g.
300 x 100 x 3

(3 for 3 color channels RGB)



What the computer sees

Challenges: Viewpoint Variation



Challenges: Illumination



Challenges: Deformation






Challenges: Occlusion







Challenges: Background clutter



Challenges: Intraclass variation

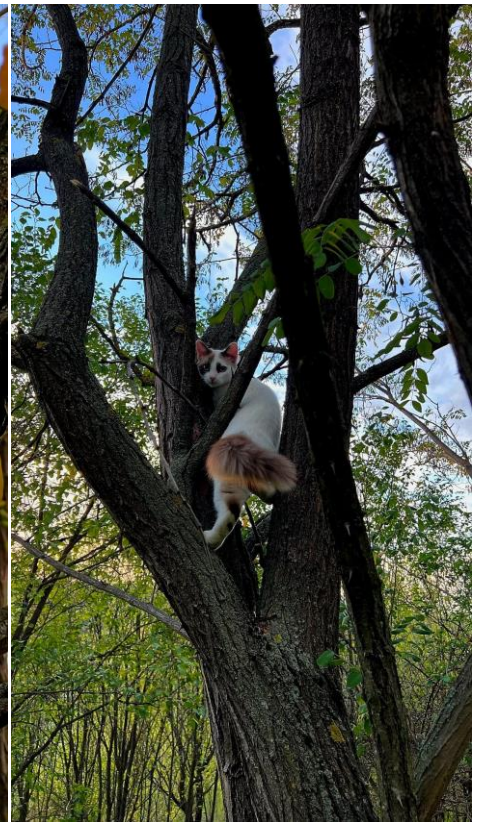


My cats ☺













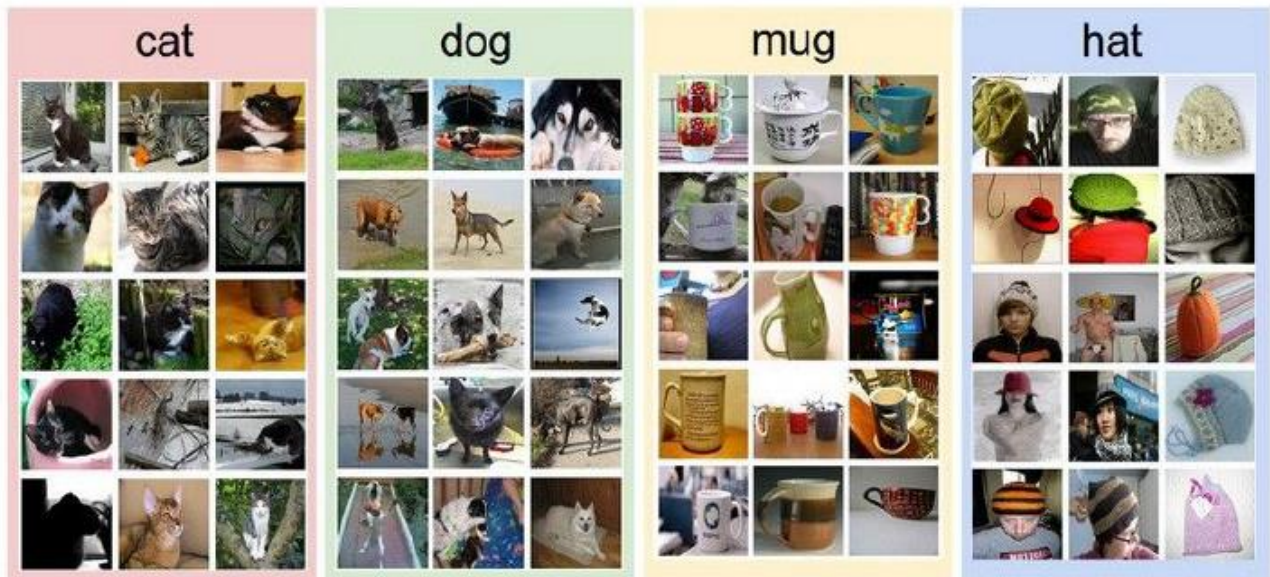
Data-driven approach:

1. Collect a dataset of images and labels
2. Use Machine Learning to train an image classifier
3. Evaluate the classifier on a withheld set of test images

```
def train(train_images, train_labels):
  # build a model for images -> labels...
  return model

def predict(model, test_images):
  # predict test_labels using the model...
  return test_labels
```

Example training set



First classifier: **Nearest Neighbor Classifier**

```
def train(train_images, train_labels):
  # build a model for images -> labels...
  return model

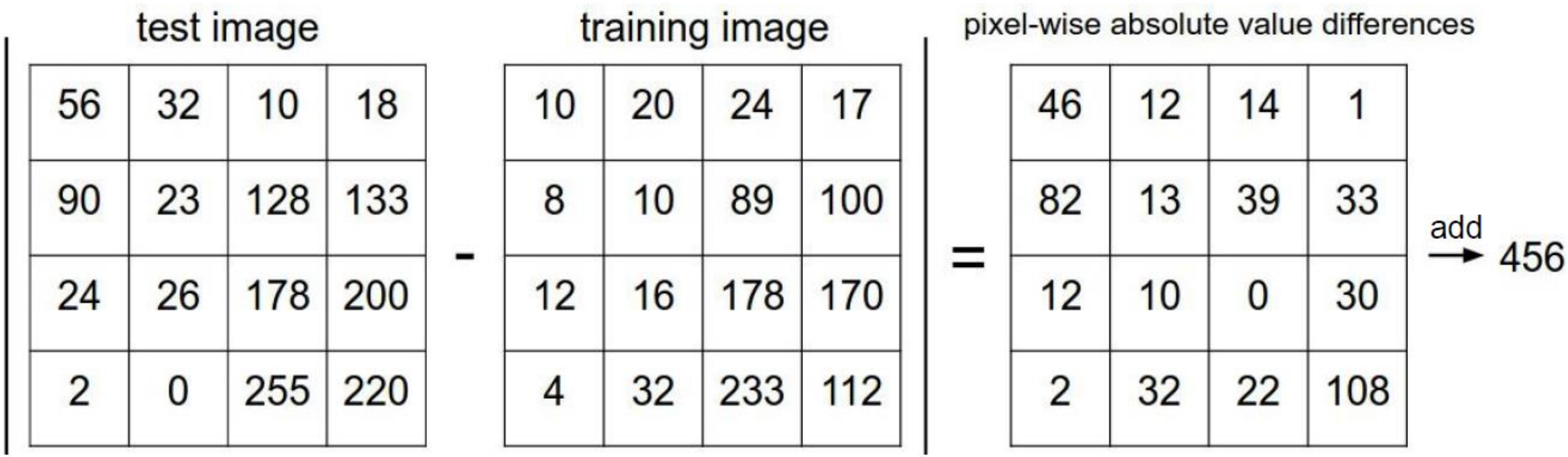
def predict(model, test_images):
  # predict test_labels using the model...
  return test_labels
```

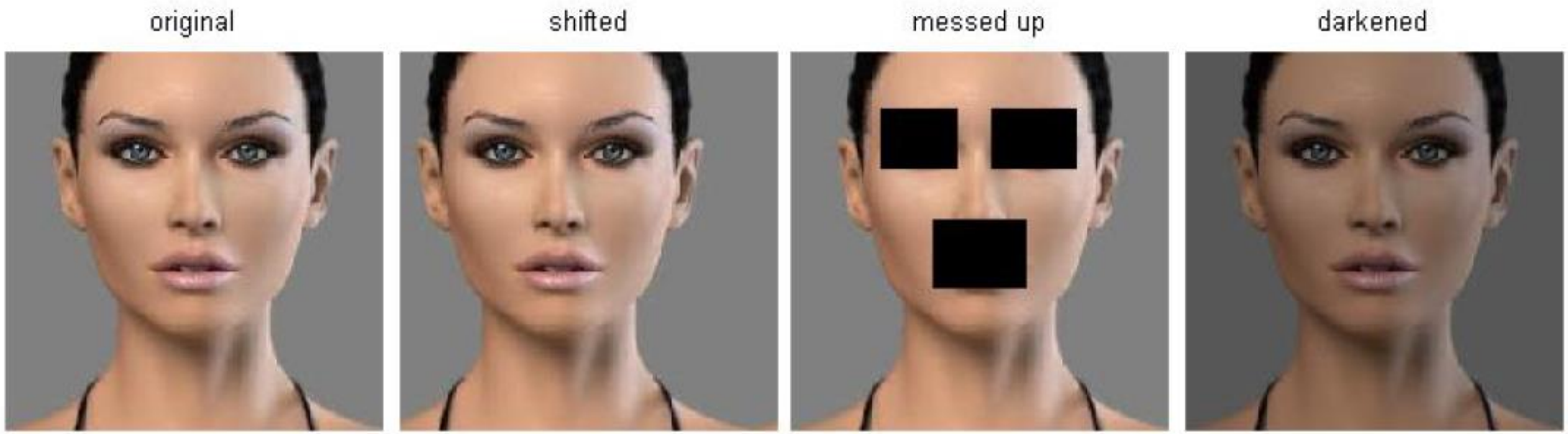
Remember all training images and their labels

Predict the label of the most similar training image

How do we compare the images? What is the **distance metric**?

L1 distance: $$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$





(all 3 images have same L2 distance to the one on the left)

Example dataset: **CIFAR-10**
10 labels
50,000 training images
10,000 test images.



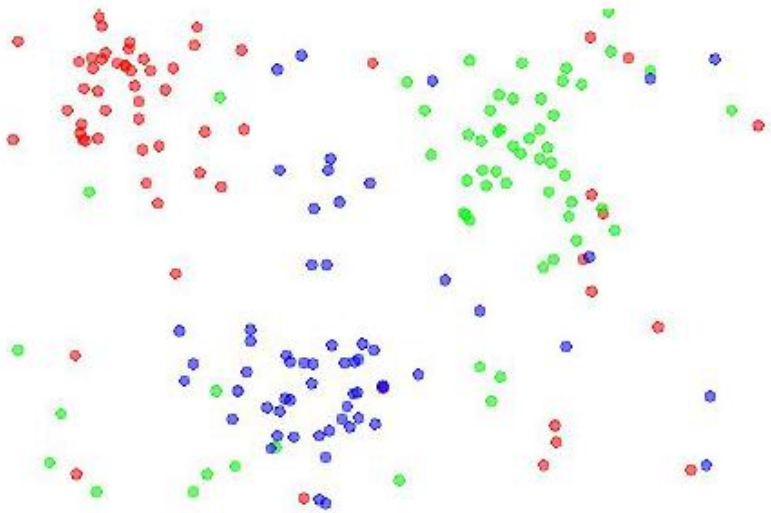
For every test image (first column), examples of nearest neighbors in rows



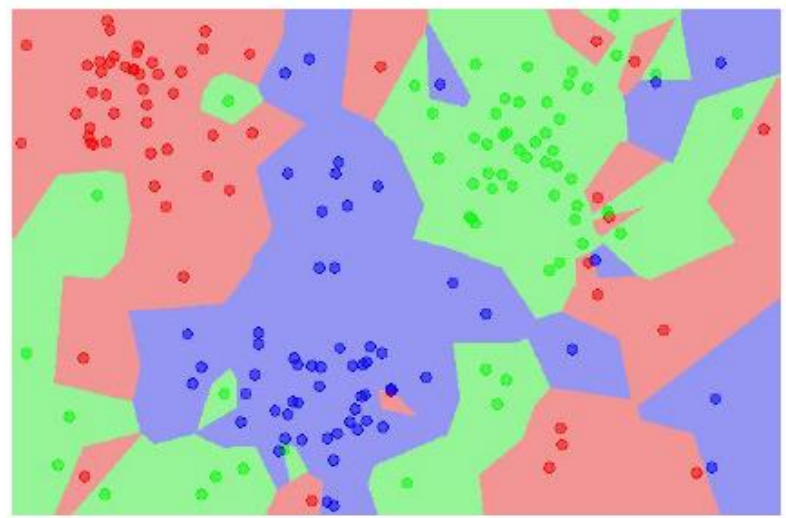
k-Nearest Neighbor

find the k nearest images, have them vote on the label

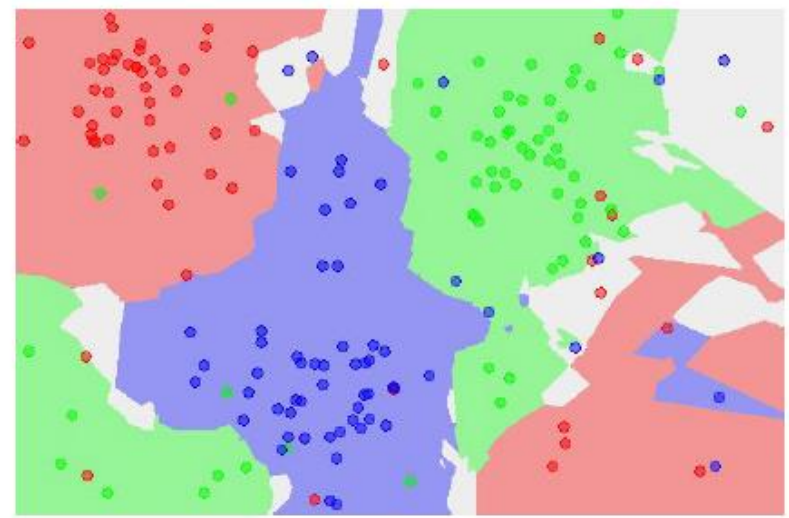
the data



NN classifier



5-NN classifier



http://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

Parametric approach



image parameters

$f(\mathbf{x},\mathbf{W})$

10 numbers, indicating class scores

[32x32x3]
array of numbers 0...1
(3072 numbers total)

Parametric approach: **Linear classifier**

$$f(x, W) = Wx$$

3072x1

(+b) 10x1

10x1

10x3072

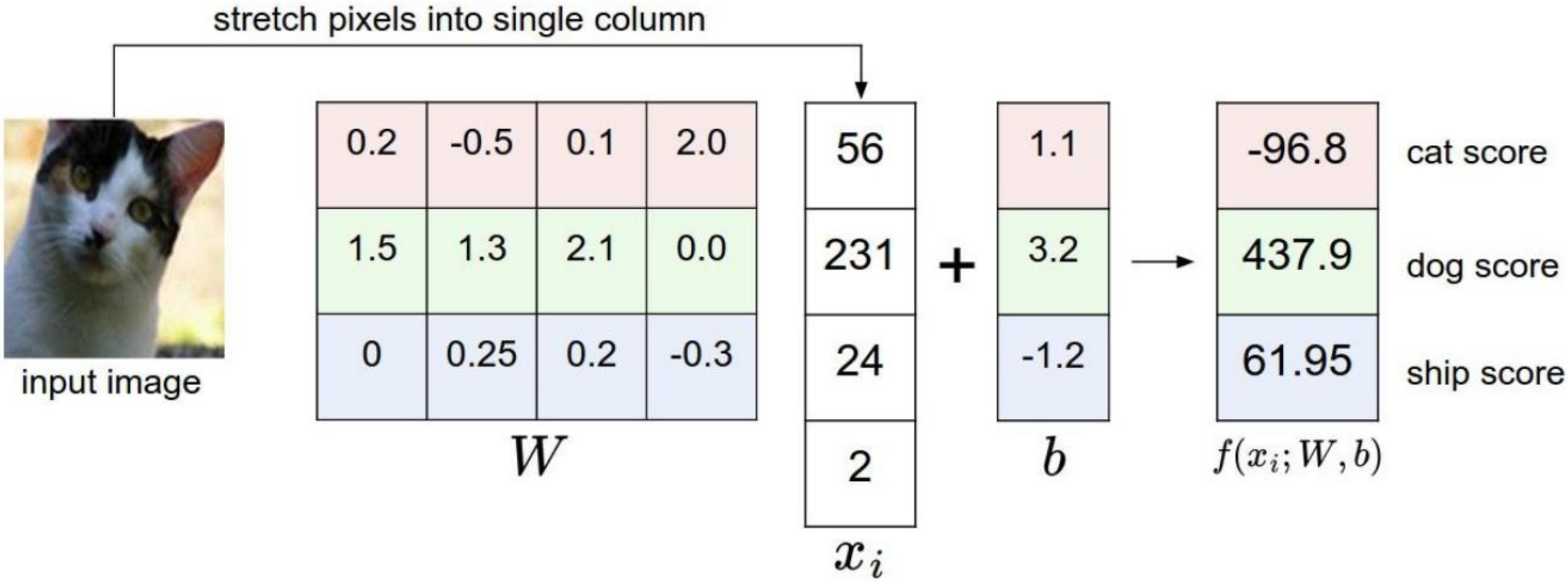


[32x32x3]
array of numbers 0...1

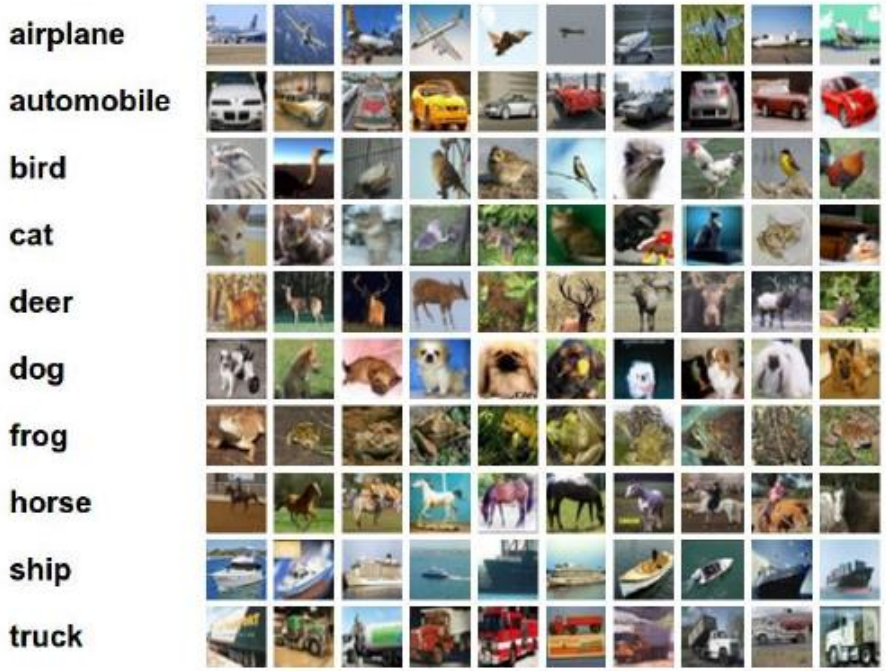
10 numbers, indicating class scores

parameters, or “weights”

Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



Interpreting a Linear Classifier

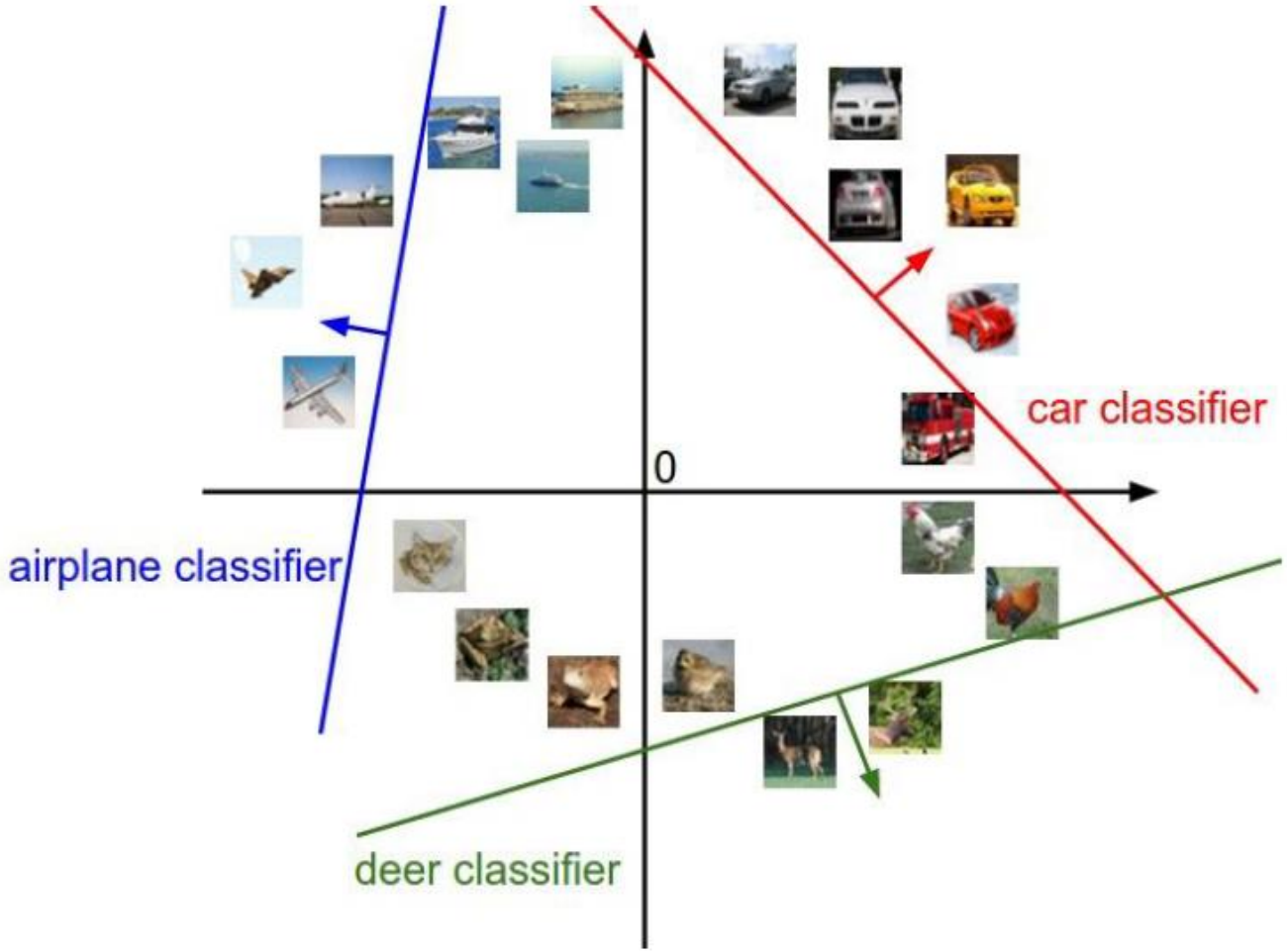


$$f(x_i, W, b) = Wx_i + b$$

Example trained weights of a linear classifier trained on CIFAR-10:



Interpreting a Linear Classifier



$$f(x_i, W, b) = Wx_i + b$$



[32x32x3]
array of numbers 0...1
(3072 numbers total)

Learning or designing an image descriptor

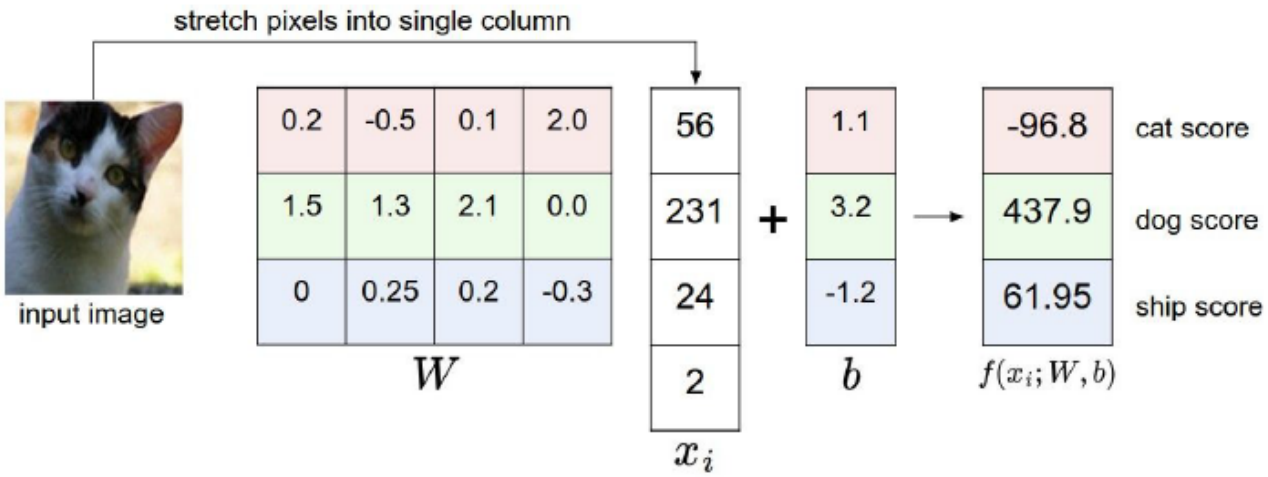
Learning the classifier parameters

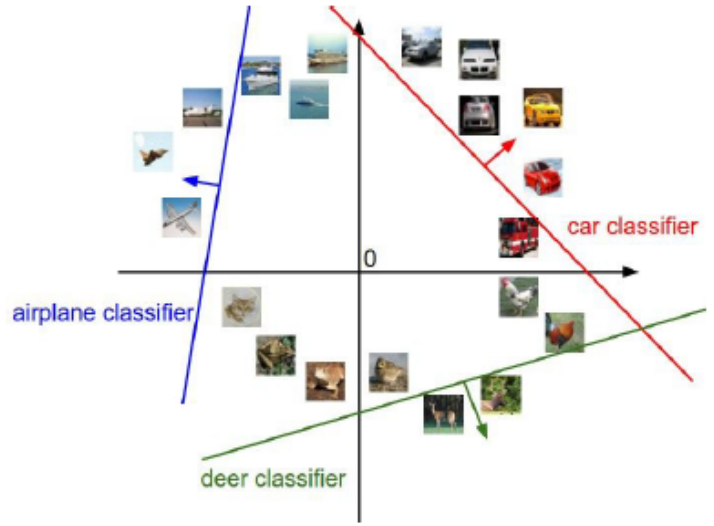


[32x32x3]
array of numbers 0...1
(3072 numbers total)

$f(x, W)$

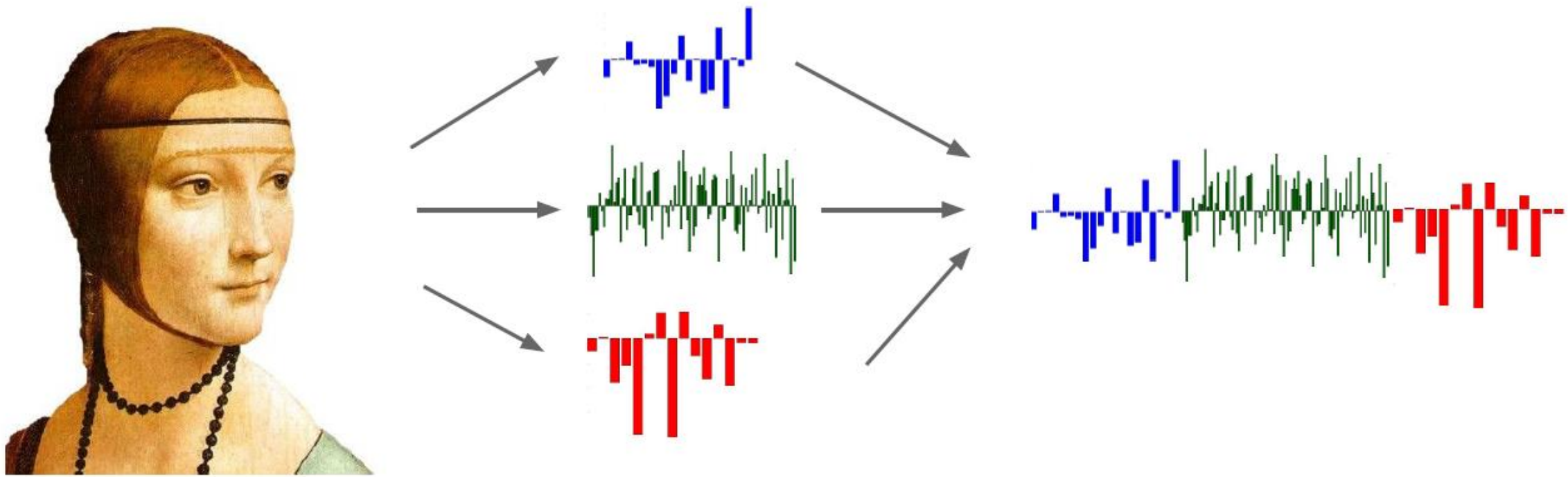
10 numbers, indicating class scores



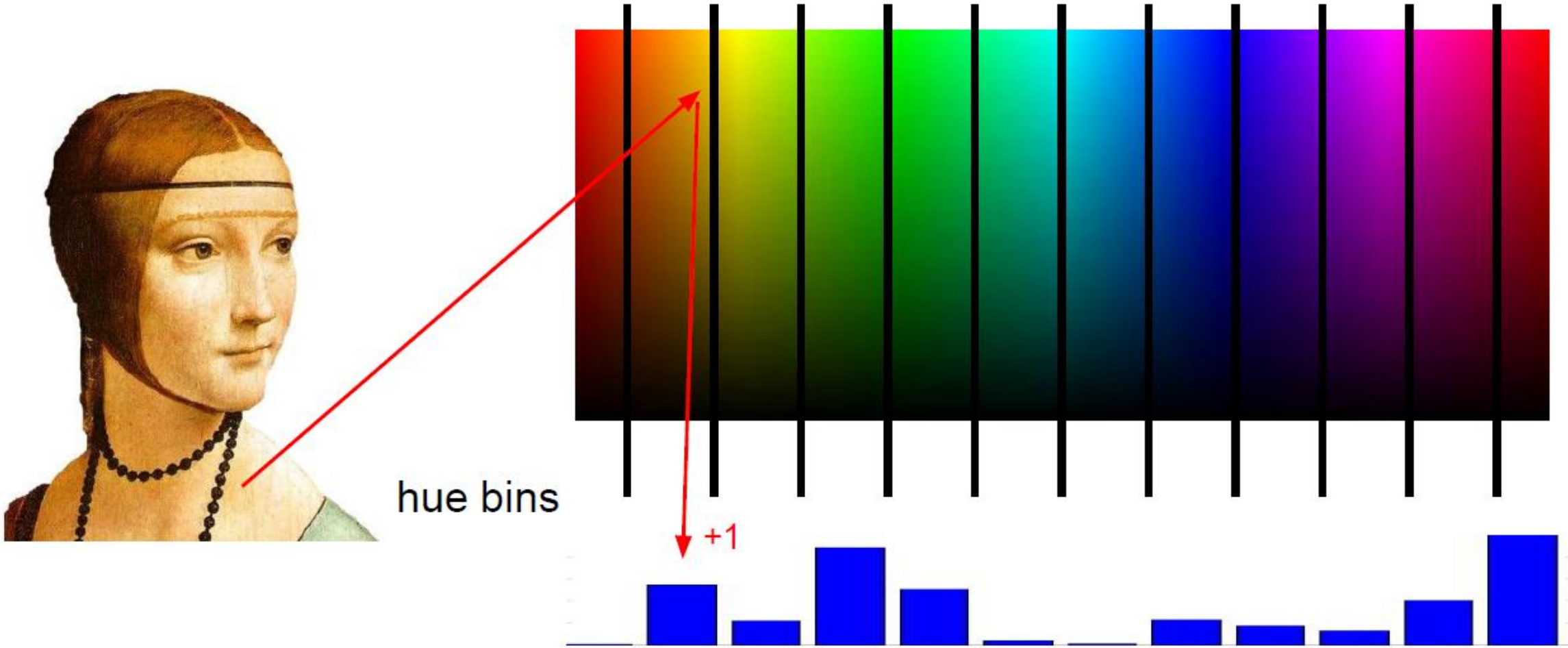




Aside: Image Features

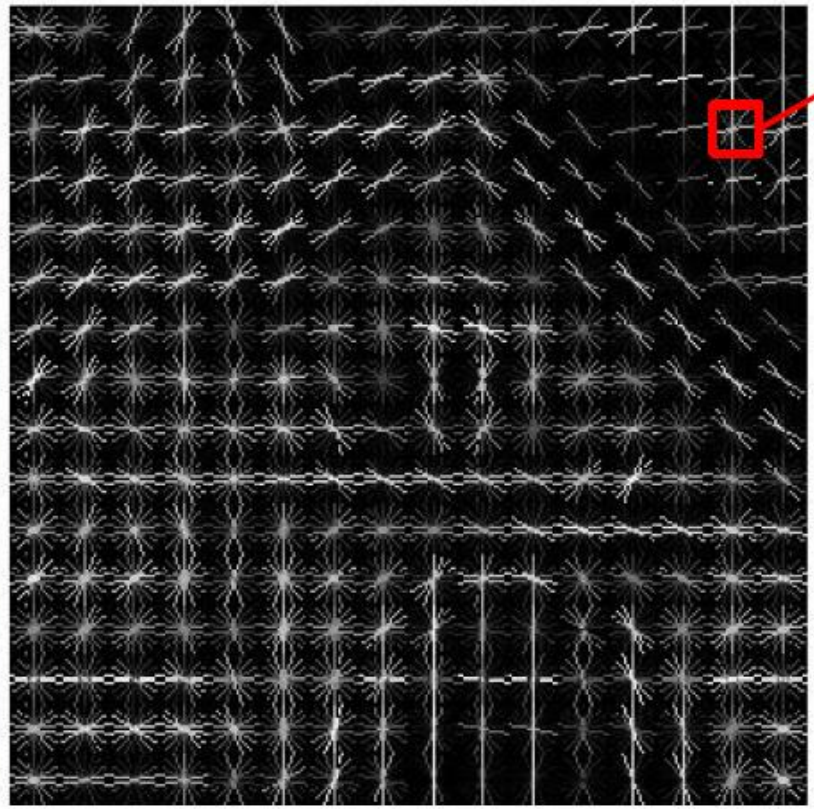


Example: Color (Hue) Histogram



Example: HOG/SIFT features



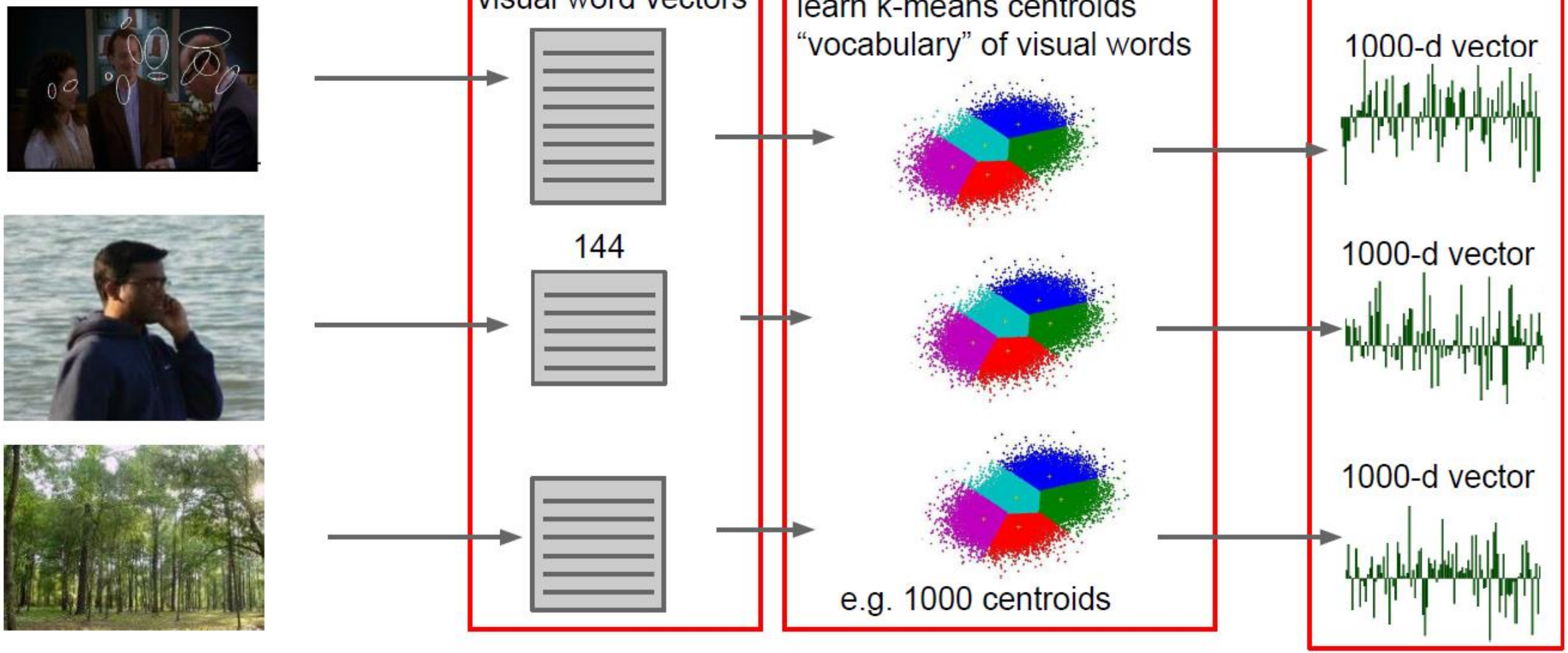


8x8 pixel region, quantize the edge orientation into 9 bins

Many more:
GIST, LBP, Texton, SSIM, ...

(image from vlfeat.org)

Example: Bag of Words



vector describing various
image statistics



[32x32x3]

Feature Extraction

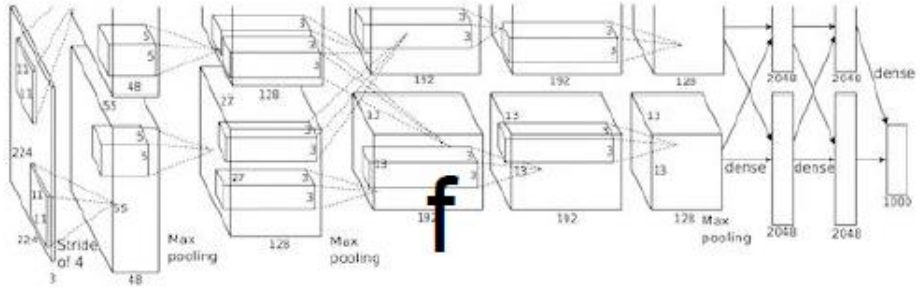
f

10 numbers, indicating class scores

training



[32x32x3]



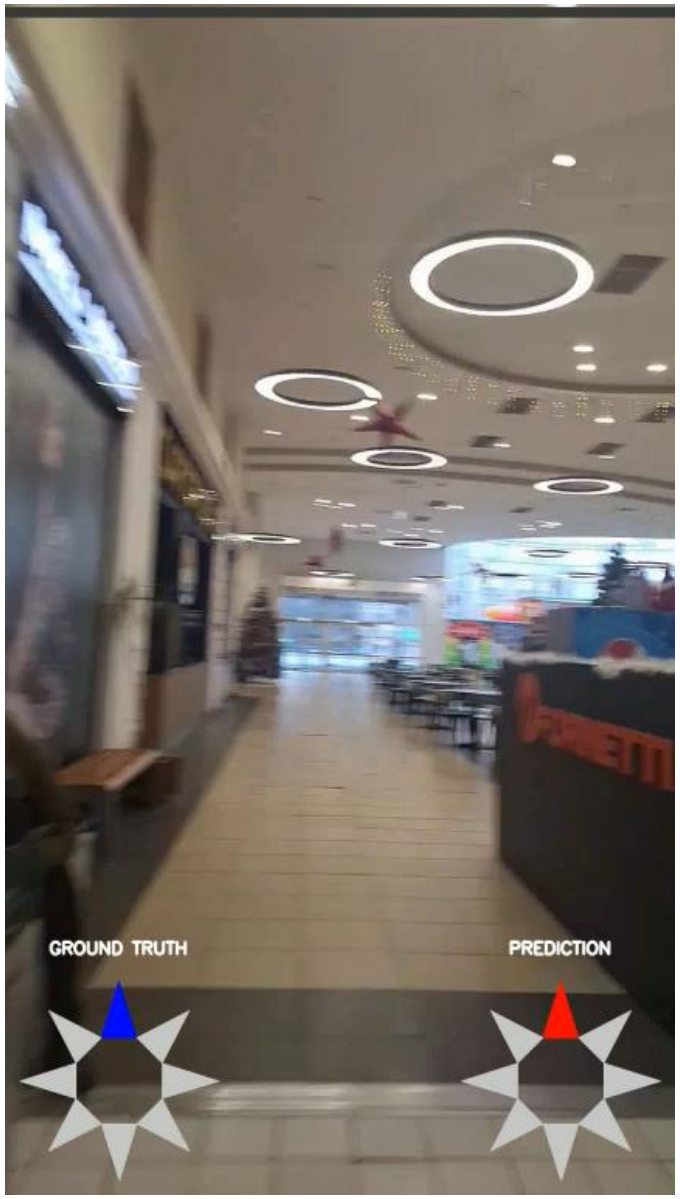
10 numbers, indicating class scores

training

Work in our lab
with Master Students

Demos





Our Art, Robotics and Human-AI Interaction at **Neo Art Connect 2024**



Team Leader:
Marius Leordeanu

PhD students:
Sebastian Mocanu
Dragos Costea
Alina Marcu

Master students:
Sebastian Nae
Mihai Barbu

Visual artist:
Cristina Lazar

Music:
Marius Leordeanu (composition, production)

Ciprian Lemnaru (production)

“Paint with Me”

Our Human-AI Interactive Art System at Kunstsillo Museum in Kristiansand (Norway)

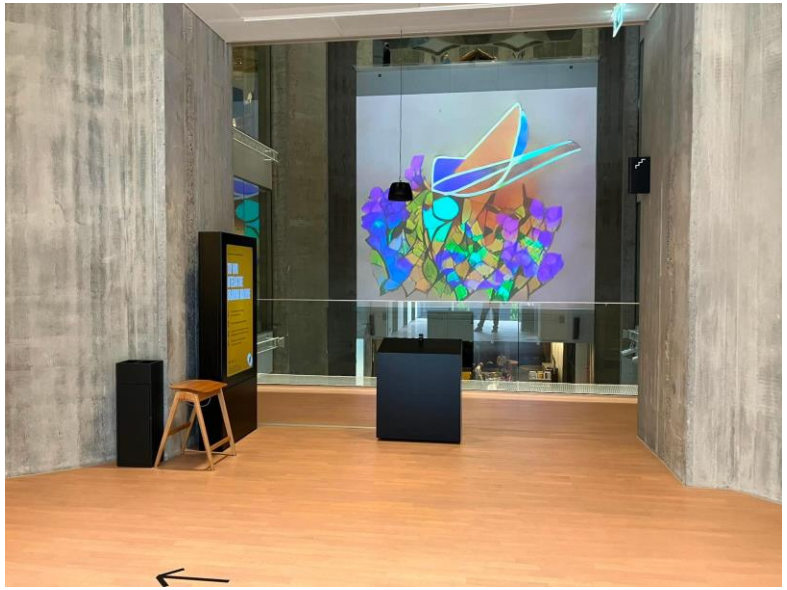




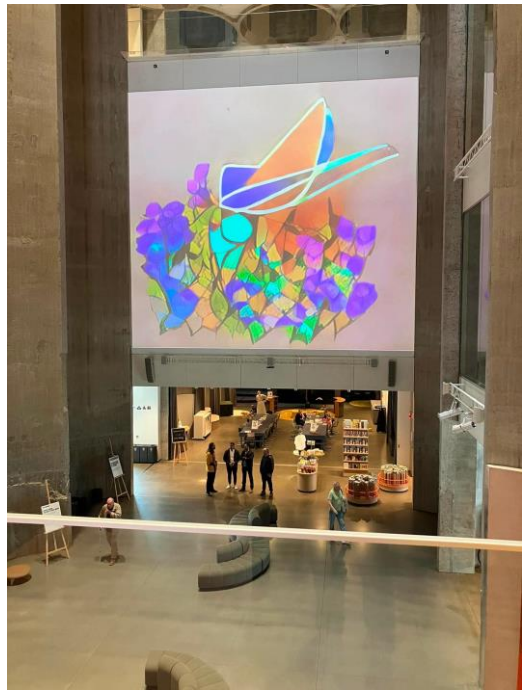














"Take me to space" – Original song composed and produced by Marius Leordeanu.
Follow me on Spotify, Apple Music, YouTube Music and others

Current work in my lab
Understand the world from flight

Transfer body language to avatars



Place and track avatars in the real scene for automatic body language learning



Targoviste – Central Park Area



"Remembering You" – Original song composed and produced by Marius Leordeanu.
Follow me on Spotify, Apple Music, YouTube Music and others

Computer Vision: Main Course Topics

Low-level Vision

Linear Filters and Edge Detection (1 lecture)
Binary Image Processing – Basic Operations (1 lecture)
Color (1 lecture)

Mid-level Vision

Model Fitting (1 lecture)
Feature extraction and Matching (1 lectures)
Clustering, Grouping and Segmentation (1 lecture)
Stereo and Geometric Computer Vision (3 lectures)

High-level Vision

Object Recognition – Deep Neural Networks (1 lecture)
Vision for Video (2 lectures)
Advanced Computer Vision Topics (1 lecture)

Invited talks by International Researchers
(1-3 lectures, depending on the time)

Grading Policy

There will be six homeworks, each worth 100 points. Late homeworks are worth 80 points each (they could be uploaded in the Late Homeworks folder).

There will be one final project worth 300 points – this will also be the final exam.

Class participation and activity will be given a maximum of 100 points.

Your final score will be the total divided by 10.

Final grade will be computed by taking in consideration the scores statistics of the entire class.

Possible Topics for Final Projects

Object Recognition in Images

Classification: ***Does this image contain a car ?***
Detection: ***Put a bounding box around the cars present in this image.***
Scene classification: ***Where was this image taken ?***
(e.g. indoors, outdoors, city, nature, classroom, in the store, seaside)

Object Recognition in Video

Object Detection in Video: Recognizing different types of objects
Finding moving objects: E.g. finding people, animals, birds, cars.
Recognizing Activities in Video: What people or animals do?
From Video to Language: Video to language translation
Semantic Segmentation in Video: Give a semantic class to each pixel.
Predicting 3D structure in Video: Predicting depth, surfaces, structure.

Many other cool computer vision applications of your choice and one

Special Project (next slide)!

Textbooks, info and contact

Szeliski, R., 2010. *Computer vision: algorithms and applications*. Springer Science & Business Media.

Forsyth, D.A. and Ponce, J., 2002. *Computer vision: a modern approach*. Prentice Hall Professional Technical Reference.

Simple and Effective: to do research use Google and Google Scholar!

My email: leordeanu@gmail.com
My website: https://sites.google.com/site/mariusleordeanu/home

Homework 1

Minimum 1 page – recommended 2-3 pages to answer and discuss the following topics and questions:

Due date: next week (7 Oct 2026)

- Why do we prefer AI models that learn an inner model of the world and how it works (from a geometric, temporal, dynamic, physics and semantics perspective) ?

- Why is it not enough to learn to predict the right class or action from directly from patterns of data, without knowhy “why” or having a model of how the world “works” and “is”?

- Argue pro or against these points…

- What is the advantage of having a model that considers many tasks and has many different interpretations of the world at the same time (it estimates geometry, movement of objects as well as their classes, together) as opposed to completely different models of different tasks ?

- Why do we prefer intelligence that learns and discovers knowledge by itself and not just supervised by a teacher or known ground truth during training?